

# Measured Quality Falls Within the Firm: Market Access and Chinese Exports

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## Abstract

Better access to a market is expected to make exporters upgrade. We show it does the opposite, and we locate where. Using the removal of tariff uncertainty for Chinese exporters to the United States in 2002, and comparing a firm's shipments of a product to the United States against its shipments of the same product elsewhere, we find that measured export quality falls by about 14 percent for heavily exposed products against barely exposed ones, and that 78% of that decline survives inside firm-product-arm cells that ship in both periods. The fall is not markup compression: the quantity response falls short of what the price cut implies by exactly the amount the measurement identity requires. We then show that this class of evidence cannot go further. Measured quality is a linear combination of the price and quantity it is inferred from, so a degraded product and a widened clientele imply the same three coefficients. We set out what would separate them.

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# 1 Introduction

When a market becomes easier to sell into, the exporters already selling there are expected to raise the quality of what they ship. The expectation rests on two well-established regularities. Richer buyers buy better goods, so a market worth reaching more of is a market that rewards quality (Fajgelbaum, Grossman, and Helpman, 2011; Verhoogen, 2008); and quality is expensive to produce, so it is the larger sellers who find it worth making, charging more for their output and paying more for their inputs (Kugler and Verhoogen, 2012).

Chinese exporters did the opposite. When the United States made China’s tariff treatment permanent in 2002, removing the annual possibility that rates would revert to their 1930 levels,<sup>1</sup> the measured quality of what incumbent Chinese firms shipped to the United States fell, and it fell relative to what those same firms shipped elsewhere. Quality is not observed in trade data, and “measured” is not a hedge: what is observed is a price and a quantity, and measured quality is the demand residual inferred from them.<sup>2</sup> It is a property of a cell’s revenue at a given price rather than of any inspected attribute of a good, and the gap between the two is this paper’s second subject rather than a caveat on its first.

The field has an answer to a falling quality measure, and it does not cover this one. Both studies that have measured quality through a Chinese liberalisation found it falling, and neither read that as a contradiction of the expectation, because both read it as a change in *who sells*. Khandelwal, Schott, and Wei (2013), on quota removal, attribute the fall to the entry and exit of firms; Feng, Li, and Swenson (2017), on the change we study, in the data we use, with the measure we use, estimate quality for entrants against exiters. A market that attracts cheaper sellers can register a lower average quality with no seller having changed anything. That reading is unavailable here. The comparison above holds the firm, the product and the destination arm fixed across the change: it is the same seller, shipping the same line, to the same place. Whatever fell, it fell inside a firm that stayed.

Showing that is harder than measuring it. Chinese export quality was moving for many reasons at once over these years, and a firm that downgraded after 2002 may simply have been a firm that was downgrading. What has to be separated is the American market changing

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<sup>1</sup>China’s access to the lower of the two US tariff schedules was renewed annually rather than granted permanently. Had a renewal failed, Chinese goods would have faced the column-2 rates of the Tariff Act of 1930, which are far higher and vary widely across products. Permanent normal trade relations, granted in 2002, removed the renewal and with it the possibility of reversion, without changing the rates actually applied. Section 2.1 sets out the institutional detail and section 3 the variation it creates.

<sup>2</sup>Within a product, arm and year, a variety selling more at a given price is assigned a higher value (Khandelwal, 2010; Hallak and Schott, 2011; Khandelwal, Schott, and Wei, 2013). Section 2.3 gives the measure and section 6.1 what it forecloses.

for a firm from the firm changing, and from the product changing everywhere. This setting permits the separation, because the policy was specific to one destination and bit unevenly across products. The same firm’s shipments of the same product to every other market therefore serve as an internal comparison group, and the bite is a tariff gap fixed before the change and outside any firm’s influence. We estimate a triple difference in exposure, arm and time on the universe of Chinese customs records for 2000 through 2014, absorbing everything that happens to a firm in a year, to a product in a year, and to each arm in a year. What is left, and what identifies the effect, is the movement in a firm’s US–non-US gap across products whose exposure differed.

Take a product whose tariff treatment would have changed a great deal had a renewal failed, and compare it with one the policy barely touched. Across that comparison — the ninetieth percentile of exposure against the tenth — measured quality on the American arm falls by 0.15 log points, about 14 percent, relative to the same firms’ shipments of the same products elsewhere. The coefficient behind that number is  $-0.349$  per unit of exposure, with price at  $-0.259$  and quantity at  $-0.098$  in the same units. We quote the percentile comparison rather than the coefficient because exposure runs from roughly zero to 0.70 across products, so a unit of it is wider than the data. The decline is not a story about which firms sell: restricting the estimate to cells that ship in both periods — the same firm, product and arm on both sides of the change — retains 78% of it, and an accounting decomposition of the same contrast puts the within-cell component at 36% of the total change while the entry term pushes the other way.

A single episode cannot settle a sign, so the design is run on a second one. The multi-fibre quotas on the same country’s exports were removed in 2005, and every post-period coefficient in that setting is negative and, once the two levers are put on a comparable scale, larger. One negative estimate can be an accident of one episode; two independent liberalisations moving the same measure the same way is harder to read that way. The two part on the margin rather than on the sign: the within-cell result is a PNTR result, for the reason section 5 gives, and section 4.4 reports what the quota estimates will and will not bear.

The cheapest explanation of a falling price is that nothing about the good changed and the seller simply took less. It is the first thing to rule out, because if it held there would be no paper. A seller who cuts a markup on an unchanged good moves buyers along a demand curve that has not moved, and at an elasticity of five a price cut of this size has to be repaid in volume: quantity must rise by 1.296 log points. It rose by nothing — the estimate is  $-0.098$ . The firm cut price hard and collected almost none of the volume that cut was supposed to buy. So the curve moved. The size of the movement is not a second finding: the 1.394 log points of missing volume are exactly four times the fall in measured

quality, which is what the measure was built to be.

What moved it admits two accounts. In the first the firm degrades the good, every buyer’s willingness to pay falls, and the price follows the cost saving down. In the second the good is unchanged and the firm sells it further down the market, so the average buyer pays less for the same thing. They cannot be told apart, and not for want of sample. Measured quality is recovered from the price and the quantity, so the three numbers a design like ours produces are two facts wearing three coats. A worse good and a cheaper clientele imply the same price fall, the same missing volume, and the same decline in the residual — at any elasticity, on any sample, in anyone’s data. That is our second contribution. The separation has to come from outside the three numbers. The model of section 6 makes the two accounts distinct objects and identifies the observable that separates them, and that observable is the product line: a firm reaching for cheaper buyers has a reason to add a cheaper variety and keep the one it has; a firm degrading its good has none.

How the two negative results this paper opened with were reached is what leaves the question open. Khandelwal, Schott, and Wei (2013)’s evidence for downgrading is not a quality estimate: it is the gap between the raw and the quality-adjusted price decline. Their direct estimate sits in their online appendix, at a 4.1 percent relative decline that is not statistically significant, and their decomposition puts the movement in entry and exit. Feng, Li, and Swenson (2017) estimate quality for entrants against exiters. Neither design could have found what we find, because neither holds the firm, the product and the arm fixed across the change, and that is the comparison these estimates are built on.

A literature has repeatedly found quality rising when a firm’s environment improves. Verhoogen (2008) and Kugler and Verhoogen (2012) find plants raising quality as they sell into richer markets, and Bastos, Silva, and Verhoogen (2018) trace the same destination-income channel through the inputs firms buy; Fan, Li, and Yeaple (2015) and Bas and Strauss-Kahn (2015) find upgrading when input tariffs fall; Fan, Li, and Yeaple (2018) find it again for Chinese exporters after WTO accession, and Ludema and Yu (2016) infer it from the price response to foreign tariff cuts. Amiti and Khandelwal (2013) find import competition raising quality growth for varieties close to the world frontier and lowering it for varieties far from it — on the exporter’s own home tariff rather than on its access abroad. Fajgelbaum, Grossman, and Helpman (2011) and Baldwin and Harrigan (2011) supply the demand-side reason to expect the pattern, and Antoniades (2015) and Fieler, Eslava, and Xu (2018) the supply-side one. Verhoogen (2023) surveys the field.

A second literature is on the policy change itself. Pierce and Schott (2016) introduce the exposure measure we use — the gap between the rate a country would pay without annual renewal and the rate it pays with it — and trace its effect on US manufacturing

employment; Handley and Limão (2017) supply the mechanism, an entry investment made under the threat of reversion, and quantify how much of China’s export boom the removal of that threat accounts for. The same variation has since been taken to the composition of Chinese exporters (Feng, Li, and Swenson, 2017), to their innovation (Liu and Ma, 2020), to their input choice (Handley, Limão, et al., 2020), to firm productivity (Brandt et al., 2017) and to prices in the United States (Amiti, Dai, et al., 2018); Handley and Limão (2022) survey it. What this literature has not asked of the episode is what happened to the good an incumbent was already shipping.

A third is the measurement of quality from trade data. Khandelwal (2010) and Hallak and Schott (2011) develop the demand-residual measure, Khandelwal, Schott, and Wei (2013) put it in the form we use, and Gervais (2015) and Piveteau and Smagghue (2019) extend and criticise it — the latter for the endogeneity of price in estimating the elasticity the measure requires. Schott (2004) and Hummels and Klenow (2005) read quality off unit values directly, Feenstra and Romalis (2014) make quality endogenous to income, and De Loecker et al. (2016) show what firm-specific markups do to inferences drawn from prices. That the residual is not quality alone is established: Hottman, Redding, and Weinstein (2016) name the object *appeal* because under CES a better good and a better-liked good are the same parameter, and Li, Li, and Zhang (2026), who observe quality directly for a panel of Chinese steel producers, show that putting the residual in its place biases the estimated cost of quality toward zero. The measure’s own authors concede the point (Khandelwal, Schott, and Wei, 2013, fn. 21). Our second contribution is addressed here rather than to the trade literature, and it is not that the residual absorbs demand — it is the consequence for the designs built on the residual, which has not been drawn out. Because the residual is a fixed linear combination of the price and the quantity it is inferred from, a design reporting all three coefficients on a common right-hand side reports two independent facts rather than three, at any elasticity and on any sample — and the mechanism questions such triples are used to settle are therefore not settled by them. The quality-adjusted price is caught by the same arithmetic: it is log revenue, rescaled.

The model in section 6 is assembled from standard parts. Buyers differ in what they will pay for quality and sort across a menu, as in Mussa and Rosen (1978); the seller’s decision is over a product line rather than a single good, which is where trade policy acted in Krishna (1990); market size governs how far down the seller reaches, as in Melitz and Ottaviano (2008); and who buys what at which income is the object Fajgelbaum and Khandelwal (2016) measure. What the model adds is the observable that separates the two accounts the identity cannot.

The paper proceeds as follows. Section 2 describes the setting, the data and the quality

measure. Section 3 sets out the design and what must be true for it. Section 4 reports the estimates, and section 5 locates them on the within-cell margin. Section 6 sets out what they cannot settle and why no estimate of this kind can, develops the model and the observable it points at, and reports how far these data can see it. Section 7 states the scope of the exercise and what would settle the question we leave open.

## 2 Data

The paper uses two liberalisations of the United States market for Chinese exports: the grant of permanent normal trade relations, effective in 2002, and the removal of the textile and apparel quotas at the start of 2005. Neither changed the tariff a firm paid at the border. Both are estimated on Chinese customs records, aggregated to cells defined by a firm, a product, a destination arm and a year, and on a measure of quality that is inferred from prices and quantities rather than observed.

### 2.1 The two shocks and their exposure measures

**Permanent normal trade relations.** Before 2002 the United States renewed China’s normal trade relations status annually, and a failed renewal would have returned Chinese goods to the column-2 tariff schedule set in 1930. Congress passed permanent normal trade relations in October 2000; it became effective on China’s accession to the World Trade Organization in December 2001 and was implemented on 1 January 2002 (Pierce and Schott, 2016). We treat 2002 as the first post-PNTR year, setting  $\text{Post}_t = 1$  for  $t \geq 2002$ .<sup>3</sup>

The two schedules were far apart. In 2000 the median product carried an applied rate of 2.4 percent against a column-2 rate of 30.0 percent, and the column-2 schedule was also the more dispersed: its first quartile and ninetieth percentile were 21 and 51 percent, against 0 and 9 for the rates actually applied. Table 1 reports both.

Permanent normal trade relations removed the renewal and with it the possibility of reversion. Applied tariffs did not move; what changed was the distribution over the tariffs a firm might face, by an amount that varies across products with the distance between the two schedules and that changed for one destination only. Handley and Limão (2017) formalise the mechanism as an entry investment made under a threat of reversion, and the product-level gap between the two schedules is the exposure measure Pierce and Schott

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<sup>3</sup>Pierce and Schott (2016) treat 2001 forward as post-PNTR. Our panel opens in 2000, so the pre-period is short under either convention; section 3.2 states what that does to the design and what we therefore do not claim.

**Table 1:** The two tariff schedules facing Chinese goods in 2000

|                          | p10 | p25  | Median | p75  | p90  |
|--------------------------|-----|------|--------|------|------|
| Rate actually applied    | 0.0 | 0.0  | 2.4    | 5.2  | 9.4  |
| Rate on a failed renewal | 0.0 | 21.0 | 30.0   | 37.2 | 51.5 |
| Exposure: the difference | 0.0 | 17.2 | 26.2   | 33.4 | 43.8 |

*Notes:* percentage points ad valorem, across the 5,033 HS6 products of the schedule file. Applied rates are reported for 4,696 products, column-2 rates for 4,361, and both for the 4,348 on which exposure is defined. The mean exposure is 25.0 points and the maximum 70.4. Rates are taken from the replication package of Handley and Limão (2017), one row per HS6–1996 code; their construction, including the treatment of specific duties and the concordance across tariff revisions, follows that source. Section D.3 traces the series back towards the statute, reports the step we could not trace, and states what we verified ourselves.

(2016) introduce. We use that variation, interacted with an arm indicator and the timing of the change, throughout.

Table 1 reports rates in percentage points, while the estimation panel carries the same quantity as a proportion, so an exposure of 0.262 is a gap of 26.2 percentage points. Across the 4,136 products of the estimation sample the median exposure is 26 points, running from near zero at the tenth percentile to 44 points at the ninetieth, with a maximum of 70. A tenth of products sit below one point. The tenth-to-ninetieth comparison we quote throughout therefore sets a product the reversion threat would have hit hard against one it would not have reached, and since no product is exposed by a full 100 points, a coefficient per unit of exposure would describe a product that does not exist.

The schedules are published at the 1996 revision of the harmonised system while the customs panel is carried at the 1988/92 revision. We map the gap across by the published concordance, averaging unweighted over the child codes of a parent that has more than one, which binds on 94 of the 4,218 products the measure covers. Among those, the within-parent standard deviation of the gap averages 0.012 against a cross-sectional standard deviation of 0.141, and weighting the children by pre-period US imports instead moves the measure by 0.0002 on average. Section D.3 reports the mapping and its coverage.

**Quota removal.** The quotas of the multi-fibre arrangement and its successor were removed in four stages, the last of which fell on 1 January 2005. China stood outside that schedule until it joined the World Trade Organization at the end of 2001, and its position then split: the goods importers had placed in the first three stages lost their quotas in early 2002, while the goods held back to the final stage lost them in 2005 (Khandelwal, Schott, and Wei, 2013). Importers used that discretion to defer what they were most protective of, so the

goods still bound in 2004 are the ones whose quotas had bitten longest. The removal was not permanent for every product: from 1 January 2006 to the end of 2008 the United States held a list of Chinese categories back under agreed import levels.<sup>4</sup>

Three records describe that regime and we use all three. The first is an annual account of the United States quotas on China, one row per quota category and year, running from 1984 to 2004 and carrying the level, the imports against it and the resulting fill rate. The second maps quota categories to tariff lines: 490 eight-digit lines across 75 categories, each line with four fill-rate columns.<sup>5</sup> The third is the Federal Register limit notices, which take over where the annual account stops: they carry 34 categories under the 2006–2008 arrangement, 18 of them ours.

Table 2 reports what the annual record holds for the five years before removal. Two things in it bear on the design. The accession reorganised the regime without reaching our products: fifty categories leave between 2001 and 2002, which is the relaxation of the goods placed in the earlier integration stages, and exactly one of them is ours. And our products are the goods held back to the final stage — the population Khandelwal, Schott, and Wei (2013) restrict their own analysis to — which here is a property of the sample rather than a restriction imposed on it: the lever is built from the last fill column, and a product with no positive fill there is dropped, so every product in the sample was still under quota at the end of the pre-period.

Exposure is built from those fill rates. A product is covered when it maps to a United States quota category, and the lever is the mean of the last fill column across a product’s quota children, which lies in  $[0, 1]$ . Products whose maximum fill rate is missing or exactly zero are dropped entirely, neither treated nor compared, their pre-period status being ambiguous: 18 products of 442, leaving 424, almost all in the textile and apparel chapters. A binary lever, equal to one where any child exceeds 0.90 — the threshold Khandelwal, Schott, and Wei (2013) take as binding — marks 181 of the 424; no estimate in this paper is run on it, and we define it because those counts are stated in its terms.

The lever is narrower than its range suggests. Across the 424 products its median is 0.85, its tenth and ninetieth percentiles are 0.65 and 0.96, and its minimum is 0.09: quotas that existed were mostly close to full. A coefficient per unit of it therefore describes a move from an empty quota to a full one, which almost no pair of products in the sample makes, and

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<sup>4</sup>Agreed in a memorandum of understanding of 8 November 2005 and implemented by the Committee for the Implementation of Textile Agreements. We take the category list from the Federal Register notices carrying the limit tables, the first of which is document 05–24175 of 16 December 2005.

<sup>5</sup>The four columns are not labelled with years in that file. The annual record is dated and covers the same categories through 2004, so we read the four as its closing years and use the last of them without relying on a calendar label.

**Table 2:** The United States quota record for China, 2000–2004

| Year | Categories  | Of which | Fill rate, our categories |                  |
|------|-------------|----------|---------------------------|------------------|
|      | under quota | ours     | Median                    | Share above 0.90 |
| 2000 | 147         | 61       | 0.95                      | 0.58             |
| 2001 | 147         | 61       | 0.92                      | 0.52             |
| 2002 | 97          | 60       | 0.96                      | 0.58             |
| 2003 | 103         | 61       | 0.95                      | 0.57             |
| 2004 | 103         | 61       | 0.87                      | 0.46             |

*Notes:* one row per year of the annual account of United States textile quotas on China. *Categories* are counted on the three-digit code and not on the record’s own label, which combines categories in one row and carries suffixes marking partial integration — a label that changes is not a quota that ends. On the label the same years show 163, 163, 108, 112 and 112. “Ours” are the 75 categories the 490 tariff lines of our concordance map into; 60 of them are under quota in every one of these five years. The fill rate is imports against the limit, and the mean is far below the median throughout, between 0.70 and 0.73, because a minority of categories sit well below their ceiling. The record ends in 2004, a year before the quotas did; for 2006–2008 the Federal Register limit notices cover 34 categories, of which 18 are ours.

section 4.1 converts to the tenth-to-ninetieth comparison, a move of 0.32, as section 4.1 does for exposure.

One feature of the lever bears on how the quota estimates should be read. It is the unweighted mean across a product’s quota children, while the definition registered for this setting is the export-value-weighted mean; these are not the same variable, and every quota estimate in this paper uses the unweighted one. The binary lever is unaffected, since it depends only on the maximum. Fill rates of this kind originate with the Office of Textile and Apparel, the source Khandelwal, Schott, and Wei (2013) name for theirs, and we did not trace our copy of the concordance to a published distribution.

## 2.2 Customs records, cells and destination arms

The data are the transaction records of the Chinese customs administration, covering all ports of entry and exit, which we obtain from the RESSET database. They run from 2000 through 2014 and carry 105,490,292 export records; imports are dropped and play no further part. A record reports the value a firm ships of an eight-digit tariff line to a destination in a year under one trade regime, together with a value per physical unit, from which the quantity is recovered. The value is in current US dollars and the quantity in the product’s own physical unit. Firms are identified by the customs party code, destinations by a country string, and products by the eight-digit code of the tariff schedule in force that year. These records are the

source behind much of the firm-level evidence on Chinese exports: Manova and Zhang (2012) establish how export prices vary across firms and destinations in them, Ahn, Khandelwal, and Wei (2011) the role of intermediaries, and Khandelwal, Schott, and Wei (2013) and Fan, Li, and Yeaple (2015) the quality responses this paper’s question descends from.

Ordinary and processing trade are pooled. Processing accounts for between 48.6 and 55.9 percent of export value in every year of the window, so the margin is not a small one, and in 2007 some 2.5 percent of firm–line–destination pairs carry both regimes, which a cell sums across. We do not split the design on the regime, and the literature gives reason to want it split: Dai, Maitra, and Yu (2016) find that processing exporters are less productive than both non-processing exporters and non-exporters, and that separating them resolves the apparent underperformance of Chinese exporters, while Manova and Yu (2016) treat the choice between the two regimes as itself an outcome.<sup>6</sup>

Destinations enter as two arms. The treated arm is the United States; the comparison arm is every other destination the same firm serves in the same product and year. We aggregate to the level the analysis uses and call it a *cell*: a firm, an HS6 product, an arm, and a year. A cell is therefore not a destination. A firm’s shipments of one product to Germany and to Brazil in the same year are one cell, not two, and the analysis never sees them apart. Within a cell we call each active HS8 code a *line*.

The comparison arm was not directly subject to the removal of US renewal risk, and it is not inert either, since a firm may reallocate capacity, inputs or product scope across destinations. What the coefficient measures is the change in a firm’s US shipments relative to its own non-US shipments of the same product, not the change relative to an untreated world; section 3.2 states what that leaves open and in which direction.

The cell price is a ratio of sums rather than an average of line-level unit values:

$$p_{fpdt} = \frac{\sum_{\ell \in fpdt} \text{value}_{\ell}}{\sum_{\ell \in fpdt} \text{quantity}_{\ell}}, \quad q_{fpdt} = \sum_{\ell \in fpdt} \text{quantity}_{\ell}, \quad (1)$$

both sums running over the transactions of the cell. Adding quantities requires that the lines share a physical unit, and they do not always. Within each product and year we keep the transactions recorded in the unit that accounts for the most export value and drop the rest — 905,967 transactions over fifteen years, 0.6 percent of export value — so that the quantity summed within a cell is measured in one unit. A transaction with no usable quantity enters neither sum, and a cell left with no usable quantity, or with non-positive value, is dropped

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<sup>6</sup>The regime classification is coarser in the later vintages of the source: the files for 2000 through 2006 distinguish fourteen to sixteen categories, those from 2007 two. The shares quoted here count as processing any category whose label contains it.

rather than carried with a price we cannot form.

Fifty-two export records in fifteen years carry a tariff code that does not resolve to a six-digit product and are removed. No record is dropped for its destination: a destination string the concordance does not recognise is assigned to the comparison arm rather than discarded. Singleton groups that the fixed effects cannot identify are removed iteratively. The estimation sample contains 6,783,323 cells, covering 26,142 firms and 4,136 products; table 12 reports the count surviving each step.

Two conventions are not visible in those counts. The firm identifier is the customs party code, normalised for its storage format only: we do not link codes across a re-registration, a merger or a rename, so a firm that changes code enters the panel as two firms. And the customs source records several US territories separately; we follow the source in treating Puerto Rico, Guam, the US Virgin Islands, American Samoa and the Northern Mariana Islands as United States destinations. Section D.7 reports what the alternative would remove and states that we do not re-estimate on it.

Table 3 reports the estimation sample. Two features of it bear on what follows. The outcomes are dispersed: the interdecile range of measured quality is close to six log points, and of log quantity close to nine, which is the variation the fixed effects and the treatment have to work within. And exposure looks different depending on what one counts. Across products its tenth percentile is near zero; across cells it is 18 percentage points, because cells are not spread evenly over products. The magnitudes in section 4.1 are stated on the product distribution, which is the wider of the two, and section 4.1 reports what the cell distribution gives instead.

The quota setting needs a third destination group and is therefore estimated on a panel of its own. The European Union held quotas on the same goods, and the panel just described pools it with the rest of the world inside one non-US cell at the level of the unit value, so the two cannot be separated after the fact. We rebuild from the same customs source with the destination carried as three groups — the United States, the European Union and the rest of the world — over 2000 through 2008. Everything else is the construction of this subsection and section 2.3 applied unchanged: the same export records, the same product mapping, the same quantity rule and modal-unit filter, the same ratio-of-sums price, the same winsorisation, and the same measure at  $\sigma = 5$ . That panel holds 3,823,874 cells, 94,323 firms and 424 products. The primary specification drops the European Union and compares the United States against the rest of the world, which leaves 3,047,220 cells once the fixed effects have dropped what they cannot identify; table 15 reports each step.

One margin is invisible in principle, in both panels. The source reports one value and one unit value per firm, line, destination, regime and year, so there is no within-line distri-

**Table 3:** Summary statistics, estimation sample

|                                                | Mean  | SD    | p10   | Median | p90   |
|------------------------------------------------|-------|-------|-------|--------|-------|
| <i>Panel A. Outcomes, across cells</i>         |       |       |       |        |       |
| Measured quality $\ln \lambda$                 | 4.18  | 2.55  | 1.46  | 3.87   | 7.24  |
| Log unit value $\ln p$                         | 1.77  | 2.37  | -0.62 | 1.39   | 4.68  |
| Log quantity $\ln q$                           | 7.86  | 3.34  | 3.18  | 8.15   | 11.92 |
| <i>Panel B. Exposure, in percentage points</i> |       |       |       |        |       |
| Across cells                                   | 31.79 | 12.20 | 17.97 | 30.01  | 49.18 |
| Across products                                | 25.15 | 14.01 | 0.16  | 26.24  | 44.02 |
| <i>Panel C. Breadth, per firm-year</i>         |       |       |       |        |       |
| Products                                       | 30.1  | 78.7  | 2     | 7      | 67    |
| Cells                                          | 34.6  | 94.4  | 2     | 8      | 75    |

*Notes:* the sample is 6,783,323 cells covering 26,142 firms, 4,136 products and 196,288 firm-years, over 2000–2014. Of those cells, 18.8 percent are on the treated arm and 81.4 percent fall in 2002 or later. Panels A and B are computed on different populations and are not comparable row to row: a cell is a firm, a product, an arm and a year, while Panel B’s second row gives one observation to each product, which is the population section 2.1 quotes. Quantities are in each product’s own physical unit, so the level of  $\ln p$  and  $\ln q$  is not comparable across products and only within-product variation is used by the design. Logs are as they enter the estimation, after the winsorisation of section 2.3.

bution of transaction prices, and a firm that adds a cheaper variety *inside* an existing line is indistinguishable in these data from one that does not.<sup>7</sup> The across-line margin is the only menu margin these panels carry; section 6.8 states what reading a line as a menu rung assumes, and section 6.1 explains why the limit matters for what the paper can conclude.

## 2.3 Quality as a demand residual

Quality is not observed. We use the demand residual that Khandelwal (2010) and Khandelwal, Schott, and Wei (2013) construct: within a product, arm and year, a variety that sells more at a given price is inferred to be of higher quality, and the residual that rationalises the observed quantity given the observed price is the measure. Under CES demand with elasticity  $\sigma$  it is

$$\ln \lambda_{fpt} = \frac{\sigma}{\sigma - 1} \ln p_{fpt} + \frac{1}{\sigma - 1} \ln q_{fpt} + \zeta_{fpt}, \quad (2)$$

<sup>7</sup>That key is unique in every year through 2007 except 2005, where twenty-six keys carry more than one row. From 2008 between 0.2 and 1.5 percent of keys do, on a dimension the file does not name; we sum over them, as we do over regimes.

where  $f$  is the firm,  $p$  the HS6 product,  $d$  the destination arm and  $t$  the year, and  $\zeta_{pdt}$  is a market-level demand shifter common to every firm selling that product to that arm in that year. We set  $\sigma = 5$  throughout and vary it in section 4.3.

We do not estimate  $\zeta_{pdt}$  separately. The outcome we construct is the first two terms of eq. (2) — at  $\sigma = 5$ , exactly  $1.25 \ln p + 0.25 \ln q$  — formed from the same winsorised price and quantity that we also report as outcomes in their own right.

Measured quality is consequently not a third independent outcome. Price and quantity are the two primitive outcomes in these data, and for a fixed elasticity and a common estimation sample measured quality is a known linear combination of their logs. Estimating eq. (3) on the three outcomes returns coefficients satisfying  $\beta_\lambda = \frac{\sigma}{\sigma-1}\beta_p + \frac{1}{\sigma-1}\beta_q$  exactly, because the outcome is that combination and least squares is linear in it. We read a quality estimate as a demand-based summary of the joint movement of prices and quantities, not as independent evidence about the physical attributes of a good. Two of the three coefficients are free and the third is arithmetic, so nothing is learned by reading them as three pieces of evidence, and a mechanism test that combines them is a test of nothing.

Objects built on top of the residual inherit this. A design need not report measured quality to be caught by it. The quality-adjusted price — the object Khandelwal, Schott, and Wei (2013) decompose alongside the raw price — is  $\ln p - \ln \lambda$ , which by eq. (2) equals  $-(\ln p + \ln q)/(\sigma - 1)$  once the absorbed terms are removed: it is log revenue, rescaled. Reporting the raw price response, the quality response and the quality-adjusted price response is therefore reporting three numbers built from two, and the difference between the first and the third is identically the second. The relation is not in dispute and is not a criticism of that paper: its own appendix states that subtracting its quality changes from its price changes yields its quality-adjusted price changes, and the published figures satisfy the identity — a 0.206 log-point raw decline against a 0.166 quality-adjusted decline, with a quality decline of 4.1 percent standing between them. What follows is not that any of those numbers is wrong. It is that the third table is implied by the other two, so a mechanism read off the three of them together rests on evidence that is not there.

Its level carries no information, and only comparisons within a market do. Because  $\zeta_{pdt}$  is never removed at construction, measured quality is interpretable only relative to other firm-product cells in the same product, arm and year. The fixed effects of eq. (3) absorb the part of that term that is additively separable into a product-by-year and an arm-by-year component. A component moving with product, arm and year jointly is not absorbed, and no specification identifying  $\beta$  can absorb it, because the regressor varies at exactly that level. That such movements are unrelated to exposure is part of what section 3.2 assumes, and the assumption is about the demand shifter as much as about prices and quantities.

The measure is a property of the cell rather than of the product. A fall in  $\ln \lambda$  says that the cell’s shipments command less revenue at a given price than they did; it does not say that any physical item is worse. We therefore call the object *measured* quality wherever it appears, and the distinction is the paper’s subject rather than a caveat on it.

Its inputs are winsorised before it is formed. Price and quantity are aggregated to the cell first, their logs are taken, and those logs are clipped to the first and ninety-ninth percentiles within a product and year. The percentiles are computed unweighted, on the cells eligible for estimation, and pooled across the two arms; a value outside its bounds is replaced by the bound, so the operation replaces values and removes no cell. Measured quality is formed from the clipped logs, so the identity above holds on the estimation sample by construction. Section D.6 reports how many values on each leg are replaced.

### 3 Empirical strategy

The comparison is internal: a firm’s shipments of one product to the United States set against its shipments of the same product elsewhere in the same year, across products the reversion threat would have hit by different amounts. Two features of the setting constrain what it can establish. The panel opens in the year the change was legislated, so there is no clean pre-period. And the comparison arm is not inert, since the quota removal of 2005 falls inside the window.

That removal is also the paper’s second setting. We run the same design on it, on a panel rebuilt from the same customs source so that the European Union, which held quotas of its own, can be held out of the comparison arm; the lever, the destination grouping and the base year are what differ, and section 4.4 sets them out where the estimates are. The equation below is the one both settings share.

#### 3.1 The estimating equation

We estimate

$$y_{fptd} = \beta (\text{gap}_p \times \text{US}_d \times \text{Post}_t) + \alpha_{ft} + \gamma_{pt} + \delta_{dt} + \varepsilon_{fptd}, \quad (3)$$

where  $f$  indexes firms,  $p$  products,  $d$  the two destination arms and  $t$  years;  $y$  is measured quality, log price or log quantity;  $\text{gap}_p$  is the product’s exposure to the policy, measured by the distance between the two tariff schedules;  $\text{US}_d$  marks the treated arm; and  $\text{Post}_t$  marks years from the change onward.

The three sets of fixed effects are what make  $\beta$  interpretable. Firm  $\times$  year effects absorb everything that happens to a firm in a year — its productivity, its credit conditions, its input

costs, and in particular its input tariff exposure, which is firm-wide and cannot survive them. Product  $\times$  year effects absorb everything that happens to a product in a year, including world prices, global demand and the policy’s own average effect on that product. Arm  $\times$  year effects absorb everything that happens to either arm in a year, including US demand, the exchange rate and US trade policy that is not product-specific.

The arm  $\times$  year effects are coarser than the word “destination” suggests. The comparison arm is an aggregate, so the effect has two groups in each year, one for the United States and one for everywhere else: a shock to a single comparison destination is absorbed through that destination’s weight in the aggregate, not as a destination effect of its own. The design neither needs nor has a separate effect for each foreign market.

The arm effects do not make the comparison arm inert. Other destinations receive their own shocks, and one of them falls inside our window: the multi-fibre quotas were removed at the start of 2005 for the United States and for other importers alike. Where a non-US destination is affected in the same direction as the United States, the triple difference understates; where it is affected in the opposite direction, it overstates. We cannot sign it, but we can bound what it reaches. Quota-covered goods are 400 of the 4,136 products in the estimation sample and carry 17 percent of its cells, and the cells that are both quota-covered and in the four years from 2005 are 4.8 percent of the sample. Section 4.1 estimates the quota removal as its own access shock.

What remains, and what  $\beta$  is estimated from, is the comparison of a firm’s shipments of the same product to the United States against its shipments of the same product elsewhere, in the same year, across products with different exposure. The comparison group is internal: the same firm and the same product, shipping somewhere other than the United States. A firm-level story, a product-level story or a destination-level story cannot produce  $\beta$ ; only something that moves the US–non-US gap differently for high- and low-exposure products can.

Shocks are correlated within firms over time and within industries across firms. The standard errors we report in the main text are clustered on the firm, and that is the number in parentheses wherever a main-text table shows one. Clustering on industry as a second dimension is the more conservative choice and roughly doubles the standard errors on the price and quality legs; section 4.3 reports what it does. It has not been computed on every sample we show, and we name the sample wherever we report it. One table departs from this and says so in its own note: the input-tariff ladder of table 5 reports the two-way form as its primary number, because the comparison it makes is across fixed-effect specifications rather than within one. Where a result turns on which form is used, we say so at the result.

## 3.2 Identification

The policy is destination-specific. It changes the terms on which Chinese firms sell to the United States and leaves their terms elsewhere unchanged, which is what permits an internal comparison group and why the estimate is not a statement about Chinese exporters in general but about the difference between where they sell.

Exposure is a product characteristic fixed before the change. It is computed from the two tariff schedules as they stood before the policy, so it does not respond to the outcomes we study, and it is constant within a product, which is why  $\text{gap}_p$  enters only through the triple interaction: its own level and its interaction with the year are absorbed by the product  $\times$  year effects.

What identifies  $\beta$  is the spread of exposure across the products a single firm ships in a single year, and almost all of that spread survives the firm  $\times$  year effects — which is what a triple difference cannot be taken on trust to have. The standard deviation of exposure is 0.122 across cells and 0.112 within a firm and year, which is 92 percent of it. The median firm ships seven products in a year and only 4 percent of firm-years carry one, so 99.8% of cells sit in a firm-year with something to be compared against. Inside those, the exposure of the most and least exposed product differs by 0.27 at the median, wider than the interquartile range of exposure across all products in table 1.

The same fact is why we do not display exposure across industries, as the industry-level studies of this policy do. Product  $\times$  year effects absorb every industry-year difference, so between-industry variation is not what identifies  $\beta$  here, and a picture of it would describe something the design discards.

The identifying assumption is that, absent the policy change, the gap between a firm's US and non-US shipments of a product would have evolved in the same way for high- and low-exposure products. It cannot be tested directly here, and what can be brought to bear on it is uneven across the two settings.

This setting cannot test pre-trends in sample. The policy was legislated in the year our panel opens, so both pre-period years lie inside the window in which firms could already anticipate the change, and only one pre-period coefficient is estimable at all. We therefore claim no parallel-trends support there and present no figure of that setting as a pre-trend test. Section 4.3 reports the year-by-year path as a description of post-period dynamics and states what the single pre-coefficient does and does not show. A longer pre-period would settle the question and this window cannot. What can be brought to bear instead is the setting that has one.

The quota setting has the pre-period this one lacks, and it is flat. Its window opens

five years before the removal, so four pre-period coefficients are estimable rather than one; from 2001 none is individually distinguishable from zero and a joint test of them does not reject, at  $p = 0.64$ . The one movement it carries is a step between 2000 and 2001, and 2000 is the year in which the allocation of the quotas themselves was being changed; section 4.4 makes that case, reports that a joint test including 2000 does reject, and reports a sensitivity analysis whose yardstick is that same step.

We take that as support for the assumption here. The two settings run the same design on the same panel and the same outcome, so a pre-period that is flat there is evidence that this design does not manufacture differential drift between a firm’s US and non-US shipments across its own products. And the threat in this setting is anticipation rather than a secular trend, which is what those years are: the quota removal became certain and dated on China’s accession in December 2001, so 2002 through 2004 are years of a known and scheduled increase in access over which that pre-period stays flat. The exposure dimensions are not the same — the two levers correlate at  $-0.02$  across the 400 products both designs contain — so what this supports is the design rather than the tariff gap itself.

## 4 Results

It does, and by enough to matter. This section reports the baseline estimates, shows that the pattern is not markup compression, and sets out what the year-by-year path does and does not establish.

### 4.1 Baseline estimates

Table 4 reports  $\beta$  from eq. (3) in both settings, on the three outcomes that the measurement identity ties together. Measured quality falls on the treated arm, relative to the same firm’s shipments of the same product elsewhere, in both. The price response is negative in both and about twice as large under quota removal. The quantity responses are where the two part: under PNTR it is small and does not exclude zero, under quota removal it is positive and does.

The coefficients are per unit of exposure, and a unit of exposure is wider than the data, so we give the magnitude as a comparison the data contain. Exposure runs from about zero at the tenth percentile of products to 0.44 at the ninetieth — a gap of 44 percentage points, in the units of section 2.1. Across that comparison the quality coefficient is 0.15 log points, about 14 percent — a heavily exposed product against one the policy barely touched. The same rescaling applies to every coefficient in this paper and we do not repeat it at each one.

The comparison is across products, and cells are not spread evenly across them, so the cell-weighted version is narrower: the tenth and ninetieth percentiles of exposure across cells are 18.0 and 49.2 percentage points (table 3), which puts the same coefficient at 0.11 log points, about 10 percent. We quote the product comparison throughout because the coefficient is a product-level object, and we report the cell-weighted figure here because it is the smaller of the two.

The sign is against the prior that better market access raises the quality a firm ships, and it is not the first negative estimate in this literature: Khandelwal, Schott, and Wei (2013) report a decline under quota removal in this same country, and Amiti and Khandelwal (2013) report falling quality growth for varieties far from the world frontier. What is new is where it lives, and that is section 5’s subject rather than this section’s.

The same design on an independent liberalisation returns the same answer. Quota removal under the multi-fibre arrangement is an independent liberalisation of the same country’s export market, and we run the same design on it, with the same quality measure and the same construction, on the panel of section 2.2 — rebuilt from the same customs source so that the European Union, which held quotas of its own, can be held out of the comparison arm. Section 2.1 defines the exposure measure and states which version of it these estimates use. Every post-period coefficient is negative and large, in the same direction as PNTR. We report that, and we also report the sensitivity analysis that prevents us from resting on it.

In comparable units the two settings are closer than their coefficients look. A quota coefficient is stated per unit of a lever whose tenth-to-ninetieth range is 0.32, against 0.44 for exposure, so the raw coefficients are not on the same scale. Rescaled on each lever’s own tenth-to-ninetieth comparison, the quota estimate is 0.21 log points, about 19 percent, against 0.15 log points and 14 percent under PNTR. The quota estimate is larger by around forty percent, not by the factor of two the coefficients suggest, and that comparison is between point estimates whose standing is not equal, for the reason section 4.4 gives.

The two settings agree where they are directly comparable. They are independent liberalisations of the same market for the same country’s exporters, run through the same design on the same outcome, and both coefficients are negative, both exclude zero under conventional inference, and the quota estimate is the larger once the two levers are put on a comparable scale. One negative estimate can be an accident of one episode; two liberalisations moving the same measure the same way is harder to read that way.

They compose it differently. Under PNTR the price leg carries the fall and the quantity response cannot be told from zero. Under quota removal the price falls by more than twice as much and quantity rises, by 0.25 log points with an interval that excludes zero — the shape released volume produces when a binding limit comes off. The identity ties the three

**Table 4:** Market access and measured export quality, two liberalisations

|                                                  | (1)<br>Permanent normal<br>trade relations | (2)<br>Quota<br>removal |
|--------------------------------------------------|--------------------------------------------|-------------------------|
| Quality $\ln \lambda$                            | −0.349**<br>(0.040)                        | −0.680**<br>(0.061)     |
| Price $\ln p$                                    | −0.259**<br>(0.035)                        | −0.594**<br>(0.049)     |
| Quantity $\ln q$                                 | −0.098<br>(0.075)                          | +0.251*<br>(0.124)      |
| Across the tenth-to-ninetieth<br>as a percentage | −0.15<br>14                                | −0.21<br>19             |
| Cells                                            | 6,783,323                                  | 3,047,220               |
| Products                                         | 4,136                                      | 424                     |
| Window                                           | 2000–2014                                  | 2000–2008               |

*Notes:* every entry is the coefficient on  $\text{access} \times \text{US} \times \text{post}$  in eq. (3), estimated on the outcome its row names, with measured quality taken at  $\sigma = 5$ . Fixed effects are  $\text{firm} \times \text{year}$ ,  $\text{HS6} \times \text{year}$  and  $\text{arm} \times \text{year}$ , the arm being the United States or everywhere else; standard errors clustered on the firm in parentheses, and section 4.3 reports what two-way clustering does to them. Within each column the three coefficients satisfy the measurement identity of section 2.3 by construction and are two independent facts rather than three. The identity holds at full precision and not at the precision displayed: under PNTR the estimates are −0.3486, −0.2593 and −0.0979 to four decimals, and  $1.25 \times (-0.2593) + 0.25 \times (-0.0979) = -0.3486$ , while the same arithmetic on the three-decimal figures returns −0.348. The two access coefficients are not on the same scale: one is per unit of a tariff-schedule gap running from zero to 0.70 across products, the other per unit of a quota fill rate whose tenth and ninetieth percentiles are 0.65 and 0.96. The middle block puts them on the comparison each lever’s own distribution contains, which is what makes them comparable. The quota column is the pooled triple difference, so that it is the same estimator as the column beside it; the event study of section 4.4 averages −0.686 over its four post-years, which is the same estimate to within six thousandths. Both columns are full samples; the continuing-cell restriction is section 5.1, and what the two settings do not share is the margin the effect sits on, which section 5 reports.

\*Significant at the five percent level.

\*\*Significant at the one percent level.

coefficients together inside each column, so each column is two facts and not three, and the difference between the columns is a difference in how one fall is made up.

Where they part is the margin. Restricted to cells that ship in both periods, the quota coefficient does not exclude zero (section 5.1), where the PNTR estimate retains 78% of its size, and the accounting decomposition that reconciles with the regression under PNTR does not reconcile here (section 5.2). The within-cell result is therefore a PNTR result and section 5 states it as one. What the two settings share is the sign and the size of the access effect; what they do not share is where it lives.

## 4.2 Alternative explanations

The most economical reading of a price fall is that the firm cut its markup on an unchanged good. That reading makes a sharp quantitative prediction, and the data reject it.

A firm that lowers its markup without changing what it ships moves buyers along an unchanged demand schedule. With  $\sigma = 5$ , the estimated price response of  $-0.259$  log points must then raise quantity by  $1.296$  log points. The estimated quantity response on the full sample is  $-0.098$ . The realised response falls short of the markup prediction by  $1.394$  log points, and by the identity of section 2.3 that shortfall is exactly  $(\sigma - 1)$  times the measured quality decline, which is  $1.394$  as well. The two routes agree, and they must: the shortfall is the quality coefficient scaled, and it does not depend on the price and quantity legs separately. They agree to thirteen significant figures, not to the three we print. Every product in this paragraph is computed from the unrounded estimates, so a reader multiplying the three-decimal coefficients printed here will recover them only to within the rounding.

The shortfall is the finding. A markup cut cannot produce it, because a markup cut leaves the schedule where it was. Something moved the schedule, and the size of the movement is what the quality coefficient records.

Two things this argument does not do. It does not rest on the quantity response being zero: on continuing cells, the sample section 5 shows the within-firm story is about, the quantity response is positive, and the same arithmetic delivers the same conclusion with different numbers, as section 2.3 sets out. Any statement about the quantity leg has to name the sample it is computed on, and we name it at every use. And it does not identify what moved the schedule: a worse good and a different clientele both move it, and section 6.1 shows that no combination of these three coefficients tells them apart.

The other economical reading is that this is a known result under another name. Chinese tariffs on imported inputs fell over the same years, and cheaper or better inputs raise measured quality (Fan, Li, and Yeaple, 2015; Bas and Strauss-Kahn, 2015). The two levers are

separated by their variation before any estimate is run. A firm’s input tariff exposure is a firm–year object: it is the same for everything that firm ships that year, to every destination. The firm  $\times$  year effects in eq. (3) absorb it exactly. What identifies  $\beta$  is the movement in a firm’s US–non-US gap across products of differing exposure, and the input lever has no such variation to give.

That absorption is also why the input effect cannot be estimated inside this specification, and why comparing the two requires taking the firm controls off in steps. Table 5 reports four rungs that differ only in how much firm heterogeneity they hold fixed. L0 is eq. (3) itself, where the lever is collinear with the firm  $\times$  year effects and no coefficient exists. L1 holds permanent firm heterogeneity fixed and is the cleanest rung at which an input effect exists at all. L2 holds firm–product heterogeneity fixed. L3 carries no firm control of any kind. Every estimate is stated in the liberalisation direction, as the percentage change in the outcome per 10 percentage point cut, so a positive entry is the direction the input-tariff literature reports; section C sets out how the lever is built.

The input effect on measured quality is a statistical zero at every rung that retains a firm control, and the positive carrying the input channel’s sign appears only where no firm control is present. L1 returns  $-0.53\%$  per 10 percentage point cut with an interval of  $[-4.13, +3.08]$ , L2 returns  $-1.58\%$  and does not exclude zero, and L3 returns  $+15.40\%$  at  $t = 8.24$  with an interval starting at  $+11.74\%$ . The evidence is the incompatibility of the intervals and not the sequence of signs: an estimate that does not exclude zero has no sign worth reading, and Panel A is not a negative that turns positive. L1’s interval lies far outside L3’s estimate and the two do not come close to meeting, so the two rungs cannot be estimating the same parameter and at least one of them carries a large bias. Section C.4 gives the reason we read it as L3: without a firm control, a firm that was upgrading for its own reasons and a firm that faced a larger input cut cannot be told apart.

The L1 interval is also tight enough to be informative about the published magnitudes, and we state which ones it excludes. Fan, Li, and Yeaple (2015) report  $+5.6\%$  to  $+8.2\%$  per 10 percentage point cut for differentiated goods, and both L1 intervals exclude that band entirely:  $[-4.13, +3.08]$  on quality and  $[-3.70, +1.61]$  on price. Bas and Strauss-Kahn (2015) report  $+0.8\%$ , which sits inside both, and we do not claim to have ruled it out: an effect of that size is one these data cannot separate from zero. One measurement wedge travels with the first comparison. FLY’s outcome is a unit value rather than a demand residual, so the measure-matched comparison is against our price column, which excludes the band on its own.

Two cells of the table point the other way. At L2 the price estimate excludes zero with the input channel’s sign reversed: a 10 percentage point cut is associated with a price  $2.49\%$

lower, on an interval of  $[-3.82, -1.17]$ , and it is the one genuine significant-to-significant reversal in the table, on the outcome that literature actually measures. And L3’s price estimate of  $+7.78\%$  lands inside FLY’s band, so the published magnitude is reproducible on these data — in the single specification that holds no firm characteristic fixed. Section C takes both further. Neither bears on the claim made here, which is only that the input channel does not generate the access result: an input effect that is wrong-signed, or that exists only without firm controls, supplies no positive one for the alternative reading to use.

Separability was tested in one direction. Every rung was estimated twice, with the input lever alone and with the access triple difference and its lower-order terms added, and the input coefficient moves by at most  $5.2\%$  of its own magnitude at every rung and on both outcomes. The converse has not been run, and section C sets out what that leaves open.

### 4.3 Dynamics and robustness

Figure 1 plots the year-by-year coefficients for all three outcomes. Read for what it establishes, the composite panel says that the response is present from the first post-period year and does not go away. The coefficient is negative in all thirteen post-period years; its interval excludes zero in twelve of them, the exception being 2004; and the first post-period year is already one of the twelve, at  $-0.103$  on an interval of  $[-0.205, -0.002]$ . From 2005 every coefficient lies between  $-0.38$  and  $-0.55$ , so none of them comes back even to the magnitudes of the first three post-period years, let alone to zero. None of this rests on the 2005 step that the rest of this subsection is about, because it holds on the years before that step as well as on the years after it.

We present the path as a description of post-period dynamics. It is not a pre-trend test, and this setting cannot provide one. PNTR was legislated in the year our panel opens, so both pre-period years lie inside the announcement window and only one pre-coefficient is estimable. That coefficient, for measured quality, is small and does not exclude zero — and it should not be read alone, because it is a cancellation: the price and quantity components at the same date are individually larger and of opposite sign, and the quantity component excludes zero. We therefore claim no pre-trend support, and we plot the components alongside the composite so that the cancellation is visible rather than hidden by it.

The post-period path carries one feature that has to be accounted for: the response does not step up at the treatment date. The first three post-years run at roughly a quarter of the pooled magnitude, and the path steps to its full size in 2005 and stays there. Four explanations are available: a delayed response; a change in the composition of cells during the window; the removal of the textile and apparel quotas in January 2005, which falls inside

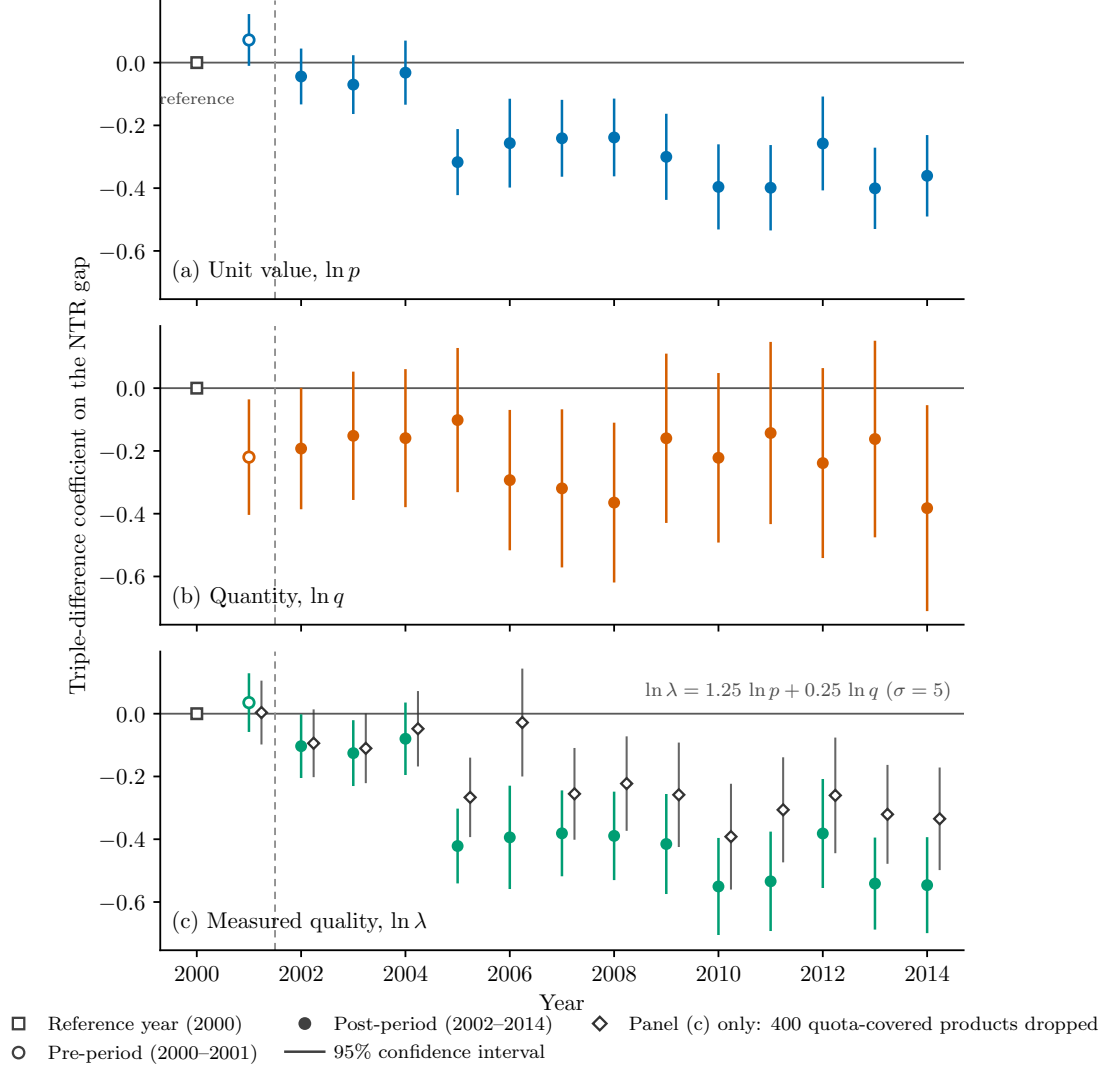
**Table 5:** Input tariffs, measured quality and price on a ladder of fixed effects

|                                | (1)<br>L0 | (2)<br>L1       | (3)<br>L2         | (4)<br>L3          |
|--------------------------------|-----------|-----------------|-------------------|--------------------|
| Measured quality $\ln \lambda$ | Absorbed  | −0.53<br>(1.84) | −1.58<br>(0.93)   | +15.40**<br>(1.87) |
| Price $\ln p$                  | Absorbed  | −1.04<br>(1.35) | −2.49**<br>(0.67) | +7.78**<br>(1.64)  |
| Firm $\times$ year             | Yes       | No              | No                | No                 |
| Firm                           | No        | Yes             | No                | No                 |
| Firm $\times$ HS6              | No        | No              | Yes               | No                 |
| HS6 $\times$ year              | Yes       | Yes             | No                | No                 |
| HS6                            | No        | No              | No                | Yes                |
| Arm $\times$ year              | Yes       | Yes             | No                | No                 |
| Year                           | No        | No              | Yes               | Yes                |
| Observations                   | —         | 6,509,511       | 5,546,425         | 6,509,702          |

*Notes:*  $\sigma = 5$ . Each entry is the percentage change in the outcome per 10 percentage point cut in the firm’s input tariff, so a positive entry is the direction the input-tariff literature reports, with the standard error in the same units beneath it. The confidence intervals are quoted in the text rather than tabulated, because what the argument rests on is where each interval sits and not whether it clears zero. Column (1) is eq. (3) itself, where the lever is collinear with the firm  $\times$  year effects and no coefficient exists. Standard errors are two-way clustered on the firm and the SIC87 industry; clustering on the firm alone gives larger  $t$ -statistics at every rung, and we report the more conservative form. The sample is the PNTR estimation sample restricted to rows with a non-missing industry code, 6,509,702 of 6,783,323 rows (95.97%), so that the two clustering schemes share identical point estimates. L1 is estimated on 191 fewer rows than L3. L2 is estimated on a smaller population than the other rungs: firm–product singletons drop it to 81.8% of the sample, and its estimates are on that restricted set. No rung is an identified causal effect of input tariffs; L0 is absorbed and L1–L3 are cross-firm by construction.

\*Significant at the five percent level.

\*\*Significant at the one percent level.



**Figure 1:** Year-by-year coefficients, PNTR. The components are plotted before the composite, on a shared vertical scale, because the composite is a fixed combination of them and the single estimable pre-period coefficient is a cancellation. Intervals are clustered on the firm. Panel (c) carries a second series, offset for legibility: the same event study with the 400 quota-covered products dropped, which is the exclusion section 4.3 reports. It is drawn on that panel alone because the price and quantity legs were not estimated on the excluded sample. The figure describes post-period dynamics; it is not a pre-trend test.

the PNTR post-period; and a specification artifact. Two of the four can be taken further with what this paper already holds.

Composition does not move. The treated arm carries 10,229 firms in 2004 and 9,558 in 2005, a fall of seven percent, which is the fall it also records in 2004 and in 2006; the cell count is flat across the same boundary. Whatever the step is, the sample is not turning over underneath it.

The quota removal is part of it and not all of it. The covered products are identified — 400 of the 4,136 products here, carrying 17 percent of the cells — so the candidate can be tested by dropping them, and the third panel of fig. 1 plots the excluded path beside the full one. Dropping them cuts the 2005 step from 0.34 to 0.22 and halves the average post-2005 coefficient, from  $-0.40$  to  $-0.19$ . The step survives the exclusion, so the quota removal does not account for it; but the products both liberalisations reached carry about half of the post-2005 decline while carrying a sixth of the cells, which is a concentration worth stating in both directions. It is consistent with two access shocks compounding on the same goods, and it means a sixth of the cells carries a disproportionate share of the later estimate.

The excluded path is also less steady than the full one, which is the reason to plot it rather than to report its averages alone. Its 2006 coefficient is  $-0.03$  with a standard error of 0.09 — a single year back at zero, with an interval wide enough to contain both the full-sample estimate for that year and nothing happening at all — before the path resumes and runs at roughly half the full-sample depth through 2014. Dropping a sixth of the cells costs precision as well as magnitude, and the year it costs most is visible in the panel.

Both candidates we can test are now tested, and the step is smaller than it was but it is not accounted for. The two that remain — a response that arrives late, and an artifact of the specification — are not tested here. The first is the less constrained of them, because a delayed response does not predict a date: with the quota-covered products dropped, the residual step still sits at 2005, on products the quota removal did not reach, and nothing in this design separates lateness from whatever else 2005 contained. Two consequences follow. The pooled estimate does not depend on what produced the step, since it averages the post-period whatever its shape. Any reading of the deeper post-2005 years as the effect growing over time does depend on it, and we do not offer that reading; the concentration reported just above is the reason.

The step is a price step. Between 2004 and 2005 the price coefficient moves from  $-0.032$  to  $-0.317$  while the quantity coefficient moves the other way, from  $-0.159$  to  $-0.102$ . Since the composite weights price at 1.25 against quantity at 0.25, what the quality panel shows at 2005 is the price movement, slightly damped. A reader looking only at the composite would attribute to quality what the components show is price, and this is the second reason

we plot all three. The quantity leg is the noisy one throughout: its standard errors run two to three times the price leg’s, and it shows no break at any date.

The estimate survives the choices a reader would want varied. Everything in this subsection is computed on the PNTR estimate. The quota setting is carried at  $\sigma = 5$  only and has no industry check; what stands behind it is the sensitivity analysis of section 4.4 and nothing further. *Elasticity*: the quality coefficient keeps its sign and significance for every  $\sigma$  in  $[3, 10]$ , a range that contains the values this literature works with — Khandelwal, Schott, and Wei (2013) assume 4;  $\sigma$  enters the measure and not the design, so the variation is mechanical and we report it for completeness. *Inference*: the standard errors in table 4 and in fig. 1 are clustered on the firm. Shocks are also correlated across firms within an industry, and clustering on that second dimension as well roughly doubles the standard errors on the price and quality legs. The quality coefficient continues to exclude zero under the two-way form, and we report the two-way estimate wherever it has been computed, naming the sample each time.

No single industry produces the result, and the one that matters most works against it. The natural worry about a product-level treatment is that one sector carries everything. We rank industries by how much each moves the relationship and re-estimate without the five that move it most, dropping one SIC87 industry at a time. The coefficient stays negative in every case and in no case becomes smaller. The most influential single industry is 3.4% of the sample, and removing it takes the estimate from  $-0.349$  to  $-0.437$ : it was damping the decline, not producing it. The other four move the estimate by between one and eight percent. A quarter of the coefficient rests on the presence of that one industry, in the direction of understatement. Only the five highest-ranked industries were re-estimated, so this is not a statement about all three hundred and fifty.

## 4.4 What the quota estimates rest on

The specification is the year-by-year form of eq. (3). We regress measured quality on the quota lever interacted with the treated-arm indicator and with each year, normalised at 2004, absorbing firm  $\times$  year, product  $\times$  year and destination-group  $\times$  year effects, and clustering on the firm. The European Union is dropped, so the destination group takes two values in estimation — the United States against the rest of the world — and the sample is the 3,047,220 cells of section 2.2. Everything that differs from the PNTR design is in that sentence: the lever, the third destination group before it is dropped, the window, and the base year.

Two exclusions and a power calculation bear on these estimates and are reported in

section [D.9](#): the products whose quotas were relaxed on accession rather than in 2005, the products the United States put back under quota from 2006, and what the joint test of the pre-coefficients could have detected. None of the three changes the reading in section [4.1](#), and neither exclusion makes the estimates more robust than the full sample does; the sensitivity analysis below is reported on those samples as well as on this one.

The pre-period is flat from 2001, and its one defect is at 2000. Across 2001 to 2004 no coefficient is individually distinguishable from zero and a joint test of them does not reject, with a  $p$ -value of 0.64; the series carries no trend, and removing the one that can be fitted makes the post-period decline larger rather than smaller. What the series does carry is a single step between 2000 and 2001. We read the parallel-trends evidence off the years from 2001, and we do not treat 2000 as the arbiter, for a reason that is about the institution rather than about the coefficient: 2000 is the year in which the allocation of the quotas themselves was being changed, and the change ended. A joint test that includes 2000 rejects, and we report that it does.

That reading is a judgement, and the case for it is institutional. From 2000 the Chinese government began experimenting with auctioning part of its quota allocation — up to thirty percent of the total for some of the goods the arrangement bound — and the authors who record this could not determine which firms or which goods were selected ([Khandelwal, Schott, and Wei, 2013](#), fn. 11). A change in how a quota licence is allocated is a change in what it costs to ship a bound good, and our exposure measure is precisely how binding a product’s quota was. The step has the shape that implies. Decomposed, it is 1.25 times a price movement of +0.267 that excludes zero, plus 0.25 times a quantity movement of +0.292 that does not: the price leg is 82 percent of it and the only significant piece. It is not concentrated in a product group, and it is an order of magnitude larger in the regression than in the raw group means, which is what a partial and selective reallocation inside bound goods would look like. It is still a candidate rather than a test, and the obstacle is the one those authors state: the goods and firms in the auctions are not recoverable. A reader who declines the argument should read the sensitivity analysis below as the verdict instead of as a bound.

We therefore keep 2000 in the pre-vector and ask, with it in, how large a post-period violation of parallel trends the estimates could tolerate, using the method of Rambachan and Roth ([2023](#)) under the restriction suited to a step rather than a slope. Keeping it in was fixed before the analysis was run, and it is the conservative choice: the yardstick that answer is measured in is that same step.

The answer is that they tolerate less than the pre-period itself displays, and what that pre-period displays is the 2000 step. Table [6](#) reports the breakdown value for each post-

year: the largest post-period violation, measured as a multiple of the largest violation visible between consecutive pre-periods, at which the robust confidence set still excludes zero. Every one of them is below one. At a multiple of one — the conventional benchmark, which allows the counterfactual trend to break after the shock no worse than it visibly did before it — no post-year set excludes zero, and neither does the set for their average.

**Table 6:** MFA post estimates and their breakdown values

| Post year | Coefficient | Conventional 95% CI | Breakdown $\bar{M}^*$ |
|-----------|-------------|---------------------|-----------------------|
| 2005      | −0.601      | [−0.786, −0.415]    | 0.90                  |
| 2006      | −0.653      | [−0.882, −0.425]    | 0.45                  |
| 2007      | −0.906      | [−1.104, −0.707]    | 0.45                  |
| 2008      | −0.582      | [−0.784, −0.380]    | 0.20                  |
| Average   | −0.686      | [−0.866, −0.505]    | 0.40                  |

*Notes:* event study at base year 2004, firm-clustered, EU destinations excluded.  $\bar{M}^*$  is the largest post-period violation of parallel trends, as a multiple of the largest violation between consecutive pre-periods, at which the robust confidence set of Rambachan and Roth (2023) still excludes zero. The restriction, both grids, the post-years and the requirement that the first pre-year enter the pre-vector were fixed before the sensitivity was computed. The same procedure on the two robustness samples of section D.9 returns 0.85, 0.40, 0.50, 0.20 and 0.40 without the products relaxed in 2002, and 0.65, 0.30, 0.40, 0.15 and 0.30 without those under safeguard from 2006. Every value on every sample lies below the benchmark.

Retaining 2000 is a choice about the yardstick rather than about the estimate, and it is the choice that costs us most. It keeps the largest pre-period movement in the denominator of  $\bar{M}$ ; excluding it would raise every breakdown value in table 6.

## 5 The margin of the decline

The decline is not a story about which firms sell. It survives inside the same firm shipping the same product to the same arm across the liberalisation, and that is the margin on which the literature has not placed it — Khandelwal, Schott, and Wei (2013) attribute their decline to entrants, and Feng, Li, and Swenson (2017) estimate quality for entrants against exiters. This section establishes that in two steps. The first restricts the estimating equation to cells that exist on both sides of the shock and carries inference. The second decomposes the aggregate change into within-cell and between-cell components and carries none.



## 5.1 Quality change within continuing cells

A firm–product–arm cell is *continuing* if it ships in both the pre-period and the post-period. Restricting eq. (3) to continuing cells removes every margin that operates through the appearance and disappearance of cells and leaves only the change within survivors.

On continuing cells the PNTR estimate is  $-0.270$ , against  $-0.349$  on the full sample: the restriction retains 78% of the effect. Price and quality both stay down and the estimate keeps a confidence interval that excludes zero under two-way clustering by firm and industry. Whatever is happening, most of it is happening inside cells that were already there.

That margin is not a new one, and its established sign is the opposite of ours. Verhoogen (2008) identifies the shift in a plant’s product mix between goods of different qualities destined for different markets as the channel linking a trade shock to labour-market outcomes, and finds upgrading: after the peso devaluation, initially more productive Mexican plants raised their export share, their wages and their rate of ISO 9000 certification relative to less productive ones. His evidence is plant-level and indirect, and he does not decompose measured quality. What our estimate adds is the same margin measured directly, on a shock that improves access rather than one that raises the return to serving a rich market, with the sign reversed.

The same restriction retains much less of the estimate in the quota setting, where the continuing-cell coefficient does not exclude zero, and that asymmetry is why this paper leans on PNTR rather than splitting its weight evenly. Under PNTR the effect is mostly inside cells that were already there; under quota removal, on the point estimates, it is mostly the appearance and disappearance of cells. The within-cell result is therefore a result about permanent normal trade relations and not about access shocks in general. One reading of the difference is that removing a threat and removing a quantity constraint act on different margins. That is a conjecture, it is not tested here, and the quota setting is in no condition to test it, for the reason section 4.4 gives.

## 5.2 Decomposing the aggregate decline

The regression answers where the effect lives conditional on a specification. An accounting decomposition answers a complementary question that no specification can: of the change in the average quality actually shipped, how much is cells changing and how much is the composition of cells changing? We apply the decomposition of Melitz and Polanec (2015) to the aggregate change in mean  $\ln \lambda$ , splitting it into a within term, a reallocation term across continuing cells, and entry and exit terms. It is an identity on a median split, so it needs no specification and carries no standard errors, and it is used here as a consistency check on

**Table 7:** Measured quality on continuing cells

|                                      | (1)<br>Permanent normal<br>trade relations | (2)<br>Quota<br>removal  |
|--------------------------------------|--------------------------------------------|--------------------------|
| Quality $\ln \lambda$                | $-0.270^{**}$<br>(0.056)                   | $-0.202$<br>(0.115)      |
| Price $\ln p$                        | $-0.285^{**}$<br>(0.044)                   | $-0.328^{**}$<br>(0.089) |
| Quantity $\ln q$                     | $+0.343^{**}$<br>(0.095)                   | $+0.833^{**}$<br>(0.207) |
| Share of the full-sample coefficient | 78                                         | 30                       |
| Cells                                | 2,649,787                                  | 1,048,028                |
| Share of the full-sample cells       | 39                                         | 34                       |

*Notes:* every entry is the coefficient on  $\text{access} \times \text{US} \times \text{post}$  in eq. (3), estimated on the outcome its row names, at  $\sigma = 5$ , with the sample restricted to cells that ship in both the pre-period and the post-period. Fixed effects, arms and clustering are those of table 4, and the two columns are the same two settings in the same order, so the tables can be read against each other. The share of the full-sample coefficient is this table's quality estimate over table 4's, computed from the unrounded figures: on the rounded ones it is a point lower. Within each column the three coefficients satisfy the measurement identity of section 2.3 and are two independent facts rather than three. The quality interval excludes zero in the first column and contains it in the second, which is why section 5.1 states the within-cell finding as a finding about permanent normal trade relations rather than about access shocks in general.

\*Significant at the five percent level.

\*\*Significant at the one percent level.

the regression rather than as a second estimate of it.

The decomposition is computed on the same contrast the regression estimates. The natural object — the change inside the US arm — is a double difference that controls for nothing outside it, so we difference it against the identical contrast in the non-US arm at a common access cut. The triple-differenced components sum to  $-0.32$ , against a regression coefficient of  $-0.35$ . That correspondence is a check and not a fit: the two objects are computed differently and are not required to agree. It holds under PNTR and not under quota removal, where the components sum to a small fraction of the regression estimate, so the decomposition is run and reported for PNTR. That is the second place the two settings part company and it points the same way as the first: the accounting of how the quota episode moves measured quality is not something these components recover.

**Table 8:** Triple-differenced decomposition of the PNTR change in mean measured quality

| Component                              | US-internal | Triple-differenced |
|----------------------------------------|-------------|--------------------|
| Within, continuing cells               | $-0.16$     | $-0.11$            |
| Reallocation, continuing cells         | $-0.75$     | $-0.22$            |
| Entry                                  | $+0.25$     | $+0.05$            |
| Exit                                   | $-0.01$     | $-0.03$            |
| Aggregate change in mean $\ln \lambda$ | $-0.67$     | $-0.32$            |

*Notes:* value-weighted,  $\sigma = 5$ . Components are additive by construction; the entries shown are rounded to two decimals and sum to the aggregate up to that rounding. The US-internal column is the high-minus-low access contrast inside the US arm; the triple-differenced column differences that against the identical contrast in the non-US arm at a common access cut. The decomposition is an accounting identity on a median split and carries no standard errors.

Table 8 carries the substantive result of this section. Reallocation was mostly a shock common to both arms; the within margin was not. Differencing against the non-US arm cuts the reallocation term from  $-0.75$  to  $-0.22$  and the within term only from  $-0.16$  to  $-0.11$ . The within-cell component is therefore 36% of the triple-differenced decline, against 24% of the US-internal one: the margin this paper is about strengthens when the comparison arm absorbs what was common to both.

That split has a literature, and both of our terms land where it would put them. Pavcnik (2002) decomposes the aggregate productivity gain following Chile’s liberalisation in the manner of Olley and Pakes: of a rise of 19 percent over seven years, 6.6 points come from productivity rising within plants and 12.7 from the reallocation of resources toward more efficient ones. Reallocation is about twice the within term there, which is the ordering our value weighting returns as well.

Trefler (2004) separates the two margins by the level at which he estimates rather than

by an accounting identity, and finds that they answer to different shocks. The Canadian tariff cuts raised industry-level labour productivity by 15 percent through the contraction of low-productivity plants; the United States’ tariff cuts — an improvement in access, which is the shock this paper studies — raised labour productivity within plants by 14 percent. Bloom, Draca, and Van Reenen (2016) find the same emphasis on the within margin for Chinese import competition, where the within-firm component of the patenting growth they attribute to Chinese imports is a little under half and reallocation a little over, and where for information technology and productivity the within component is the larger of the two.

Our shock is an improvement in access and our effect is inside continuing cells, which is the margin Trefler’s access shock acts on. The difference is the sign, and the sign is this paper’s subject. On which of our two terms is larger the weighting decides, as section 5.2 sets out; what does not depend on the weighting is that the within term is the more robust of the two to the comparison arm.

The entry term is positive in both columns. Cells that appear during the episode ship higher-quality varieties than the incumbent average, which is the pattern Feng, Li, and Swenson (2017) find for new exporters under this same policy change, and it pushes the aggregate up rather than down. The decline is therefore not an entry composition effect, and the entrant quality those accounts document is real in our data as well. Differenced against the non-US arm the entry term is small, +0.05 against an aggregate of  $-0.32$ , so it offsets rather than drives.

Reallocation is the largest single term, and this paper’s claim is about the within margin: the size of the reallocation term is reported and what generates it is not claimed, because the model in section 6 does not produce it. And the result does not rest on the weighting. Value-weighted, reallocation is about twice the within term; count-weighted, on the same sample, the within term is the larger of the two. What is the same under both is the sign of the continuing-cell total, which is negative either way; it is the entry and exit terms, the margin the model excludes, that make the all-cell aggregate weighting-dependent.<sup>8</sup>

The within term was the subject of a registered directional prediction, and the prediction failed in the informative direction: the within margin survives the rest-of-world control rather than washing out as a shock common to both arms. No directional prediction was registered for the reallocation or entry terms, which are reported as new information rather than as

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<sup>8</sup>Including entrants and exiters, the four components are  $-0.115$ ,  $-0.220$ ,  $+0.049$  and  $-0.034$  value-weighted, summing to  $-0.320$ ; and  $-0.038$ ,  $-0.006$ ,  $+0.243$  and  $-0.023$  count-weighted, summing to  $+0.177$ . Both aggregates are computed from the unrounded components; the figures displayed here are rounded and sum to them only up to that rounding. The all-cell aggregate changes sign between the two weightings, driven by the entry term. Both weightings are reported in the source and neither is taken there as the answer.

tests.

## 6 The model

The identity leaves the mechanism open, so this section supplies one that produces the fall without any good becoming worse in any buyer’s eyes: better access lets a seller reach further down the distribution of buyers, and the buyers it reaches are the ones on whom quality does not pay. The cost elasticity  $\eta$  is both the threshold that sorts buyers within a seller’s own menu and the boundary at which reaching further turns the quality choice downward, which is what links the two margins the section works with. It closes on what the panel can and cannot see of the second one.

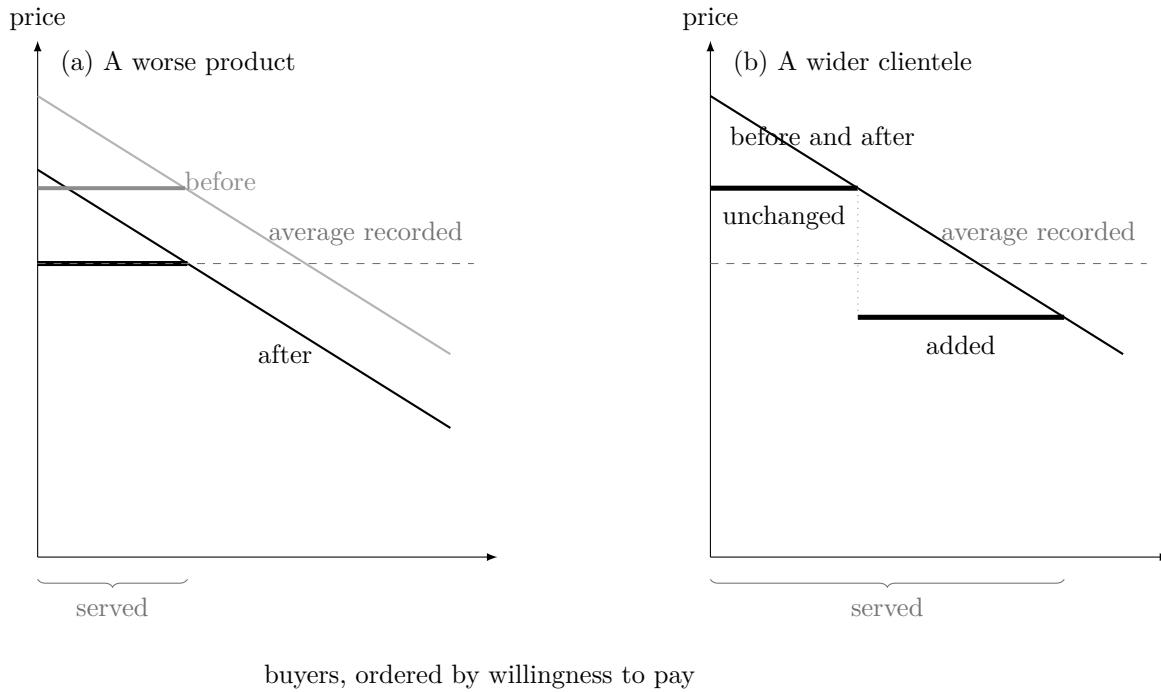
### 6.1 The identification problem

The restriction that sends this paper to a model is a property of the measure rather than of our sample, and it binds any study that infers quality from prices and quantities this way. Section 2.3 sets out the identity: measured quality is a fixed combination of the price and the quantity, so the triple difference returns two independent facts in three coefficients. Section 4.2 shows what that arithmetic does establish — the quantity response falls short of what a markup cut requires, so the demand schedule moved. It does not establish what moved it.

Both candidate mechanisms produce the same three numbers. Suppose the firm lowers the appeal of what it ships. The willingness to pay of every buyer falls, the schedule shifts inward, and the cost saving is passed through as a lower price. Now suppose instead that the firm ships the same good to a different set of buyers, reaching further down the willingness-to-pay distribution. The average buyer pays less for the same good, the measured residual falls, and the price recorded for the cell falls with the mix. In the first case the good is worse. In the second it is not. Both produce a price fall, a quantity response short of the elasticity, and a decline in measured quality of exactly the size the identity requires. The triple difference returns the same three numbers under both, at any  $\sigma$ , on any sample.

The separation therefore has to come from outside the triple. Because eq. (2) ties the three outcomes together, what distinguishes the two accounts has to come from outside it, and there are two kinds of thing it can be.

The first is a restriction on demand itself. The reason the two accounts coincide is CES: a good that is better and a good that is better liked raise revenue in exactly the same way, which is why Hottman, Redding, and Weinstein (2016) name the object *appeal* rather than



**Figure 3:** The two accounts of a fall in measured quality. The downward-sloping line is what buyers will pay, buyers ordered from the most to the least willing; the heavy horizontal segments are the prices the seller posts. In (a) the seller makes the good worse, so the line itself falls and the single price with it. In (b) the good is unchanged, so the line is where it was, the price the seller already charged is where it was, and a lower price is added underneath it for buyers who were not served before. The average price the data record is the same in both panels, and so are the quantity and the residual built from them: that is what section 2.3's identity forecloses. What differs is the number of prices the seller posts and whether the original survives, which is the observable section 6.7 turns on. The figure is a schematic and carries no data; the horizontal extents are illustrative.

quality and treat quality and taste as one parameter. Separating them therefore requires a demand system in which they act differently, and Di Comite, Thisse, and Vandebussche (2014) supply one: under quadratic preferences, quality shifts a variety’s demand curve in parallel while taste rotates it, so the two enter as distinct parameters rather than as one. That route is closed here. Its identification runs through the firm’s markup — the taste parameter is a variety’s markup divided by its quantity — and so requires costs at the firm–product level. Di Comite et al. do not have them either: they test their model’s fit against the alternatives rather than estimate its parameters, and record full identification as requiring a much richer dataset than export records. The obstacle is not that nobody has written down a demand system that separates the two. It is that separating them takes data on costs, and the coefficients this design reports are not that.

The second is an observable the identity does not determine, and two are available in principle. One is the distribution of transaction prices within a cell: a worse good shifts the whole distribution down, while a wider clientele adds mass at the bottom and leaves the top where it was. Our customs source is aggregated to one record per firm, line, destination and year, so no within-line price distribution exists in it, and this route is closed here rather than in principle. The other is the firm’s product line itself, the difference fig. 3 draws: a firm that widens its clientele has a reason to add a cheaper variety and keep the existing one, while a firm that degrades its good has none. The model set out below makes these two accounts distinct objects rather than two descriptions of one, and section 6.7 shows that the second observable separates them. Section 6.8 states what that separation costs: it holds under a maintained mapping from a model variety to a tariff line, and that mapping is assumed rather than identified.

## 6.2 The mechanism

Buyers differ in how much they are willing to pay for a better version of the same good. A seller choosing what to ship chooses two things at once: how good to make it, and how far down that distribution to sell. The two choices are not separable, and the link between them is what this model is about.

Making the good better costs more per unit. That cost is worth bearing for a buyer who values quality highly and not for a buyer who does not, so there is a buyer type at which the two exactly offset. Above that type, a buyer pays for quality more than quality costs; below it, less. The type is a property of the cost function rather than of any buyer, and it is the same number for every seller in the industry.

That structure is Mussa and Rosen (1978)’s. A seller there offers what they call a product

line, a quality-differentiated spectrum of one generic good; buyers purchase one unit, differ in what they will pay for quality, and self-select along a posted price–quality schedule, with marginal cost increasing in quality. Our menu, our rule that a buyer takes one variety, and our sorting boundary are theirs. What differs is the force: in Mussa and Rosen the seller distorts quality downward to screen a fixed population of buyers, whereas here quality falls because the population changes. Their mechanism operates at a given clientele; ours operates through the clientele.

Now improve a seller’s access to a market. The seller finds it worth serving buyers it previously skipped, and those buyers are at the bottom of the distribution. But a buyer below the offsetting type is only worth serving at a lower quality — serve him at the old quality and he does not cover what that quality costs. Reaching further down therefore drags quality down with it. The seller has not become less capable and faces no higher cost; it has changed who it is selling to, and the quality it ships follows.

That is the whole mechanism. What the rest of this section adds is, first, that the same force operates on a second margin the seller has — it can keep its existing variety and add a cheaper one rather than degrade what it has — and second, that which of the two it does is the observable that separates a worse product from a wider clientele.

### 6.3 Environment

The model’s agent is the same object the empirical sections call a firm; we call it a *seller* here to mark that its choices are the model’s rather than the data’s. Its cost elasticity is written  $\eta$ , reserving  $\beta$  for the regression coefficient of eq. (3).

A continuum of sellers supplies a market populated by buyers who differ in  $\theta$ , their marginal valuation of quality, distributed on  $[\underline{\theta}, \bar{\theta}]$ . A seller chooses a quality  $a \geq 1$  and a price for each variety it offers, and chooses which buyers to maintain a relationship with, each relationship costing a flow  $k$ . Quality raises marginal cost with elasticity  $\eta$ , and each variety offered pays a fixed cost. Demand across sellers is constant-elasticity with elasticity  $\sigma$ ; within a seller, a buyer takes the single variety that is cheapest in quality-adjusted terms. Section A states the conditions in full and marks which are maintained.

Each of those pieces is standard and none is ours. The industry side — a continuum of sellers of heterogeneous productivity facing constant-elasticity demand and paying a fixed cost per product — is Melitz (2003). Buyers of different types sorting across qualities is the structure Fajgelbaum, Grossman, and Helpman (2011) bring to trade. That quality raises marginal cost is the regularity Kugler and Verhoogen (2012) document at the plant level, where more capable plants buy higher-quality inputs, pay more for them, and sell

higher-quality output at higher prices. What is assembled rather than borrowed is the joint determination of reach and quality through one elasticity, which is what section 6.4 sets out.

A buyer of type  $\theta$  has outlay  $Y(\theta)$ , of which a share  $\psi(\theta)$  reaches the seller's source group, whose price index at  $\theta$  is  $P(\theta)$ . The three enter only through their product,

$$m(\theta) \equiv Y(\theta) \psi(\theta) P(\theta)^{\sigma-1} > 0, \quad (4)$$

so a buyer matters to the seller in proportion to  $m(\theta)$  and to how much that buyer's valuation magnifies quality. Writing  $g$  for the density of  $\theta$ , the weight a served buyer carries in the seller's quality decision is

$$\omega_a(\theta) \equiv m(\theta) a^{(\sigma-1)\theta} g(\theta), \quad (5)$$

which is the  $\omega_a$  of eq. (7) below.

Nature draws productivities and sets the policy state. Each seller then chooses its reach and its menu *together*, taking the price indices and its rivals' offers as given. Buyers then allocate their spending across sellers and, within a seller, buy the one variety that is cheapest in quality-adjusted terms. Reach and menu are one stage rather than two, for the reason section 6.4 gives: the composition of the clientele determines the return to quality, and the quality determines which buyers are worth reaching. A recursive timing would beg the question the model exists to answer.

Access enters as  $\rho$ , the probability that the market's tariff treatment is not reverted. The reverted state is the same environment at a different tariff, so the value of a relationship in that state is a fixed fraction of its value in the good state. Because that fraction does not depend on the buyer's type or on the quality shipped, the whole of the policy enters every marginal condition as a single scalar weight on variable profit. With  $t_1$  the column-one tariff and  $t_2$  the column-two rate that applies on reversion, that fraction is  $\chi = [(1+t_2)/(1+t_1)]^{-\sigma} \in (0, 1)$ , and the weight is

$$\varpi(\rho) \equiv \rho + (1 - \rho) \chi, \quad \varpi(0) = \chi, \quad \varpi(1) = 1, \quad (6)$$

strictly increasing in  $\rho$ . It is the  $\varpi(\rho)$  that appears in eq. (9).

Treating the policy as a probability that treatment does not revert follows Handley and Limão (2017), and the difference from their model is worth naming. They carry uncertainty through sunk entry costs, which generates an option value of waiting; the threat matters because entering is irreversible. Reach here costs a flow rather than a sunk payment, so there is no option value and no discount factor, and the whole of the policy reduces to the scalar weight above. The two are not competing accounts of the same margin: theirs governs

whether a seller is in the market, ours governs how far down it sells once it is. The invariance of  $\chi$  is in  $\theta$  and  $a$  only: it is a ratio of gross tariff factors and may differ across goods carrying different schedules, so nothing here makes  $\varpi$  common across goods. Granting permanence raises the weight; it does not change any other object in the model.

## 6.4 The sorting boundary

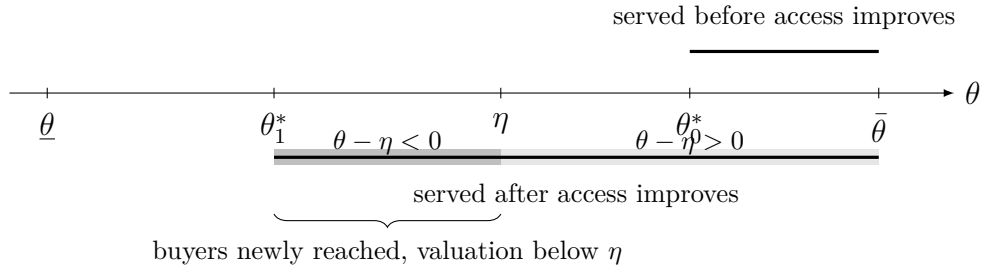
Write  $\Psi(S; a)$  for the marginal value of quality to a seller serving the set of buyers  $S$  at quality  $a$ . Under the conditions in section A,

$$\Psi(S; a) = \int_S (\theta - \eta) \omega_a(\theta) d\theta, \quad (7)$$

where  $\omega_a$  is eq. (5). Because the served set is the upper interval  $[\theta^*, \bar{\theta}]$ , moving the cutoff changes the value of quality at the rate

$$\frac{\partial}{\partial \theta^*} \Psi([\theta^*, \bar{\theta}]; a) = -(\theta^* - \eta) \omega_a(\theta^*), \quad (8)$$

which is positive below  $\eta$  and negative above it, so  $\Psi$  is maximised over upper intervals at  $\theta^* = \eta$ , pointwise in  $a$ . Each served buyer contributes  $(\theta - \eta)$ : buyers above  $\eta$  pull the seller's quality up, buyers below it pull quality down, and the cost elasticity  $\eta$  is therefore also the boundary that sorts buyers within the seller's own menu. One number does both jobs, and that is what ties the two margins together.



**Figure 4:** The sorting boundary and the reach margin. Before access improves the seller serves buyers above  $\theta_0^*$ , all of whom lie above the sorting boundary  $\eta$  and so contribute positively to the return to quality. After it improves the cutoff falls to  $\theta_1^* < \eta$ , and the buyers between  $\theta_1^*$  and  $\eta$  contribute negatively: more quality does not pay on them. The figure is a schematic and carries no data.

Two consequences follow. First, a clientele that extends below  $\eta$  has a lower marginal value of quality than one that stops at  $\eta$ , at any quality level, so extending downward lowers the *return to quality*. Whether it lowers the quality the seller chooses is a separate question,

and Proposition 6.1 rather than this observation is what answers it. Second, the seller’s reach and its quality must be solved jointly, because the composition of  $S$  determines the return to  $a$  and  $a$  determines which buyers are worth serving. We solve them as a simultaneous system rather than sequentially.

## 6.5 The reach cap

The restriction below is a feasibility condition rather than a first-order condition, and survives configurations in which first-order conditions do not locate the optimum.

**Proposition 6.1** (The reach cap). *Under the conditions in section A, if a seller’s optimum serves any buyer  $\tilde{\theta} < \eta$ , then its quality satisfies  $a \leq \bar{a}(\rho, \tilde{\theta})$ , where  $\bar{a}$  is decreasing in the distance of  $\tilde{\theta}$  below  $\eta$ . Consequently, if the seller serves such a buyer after the access change and its quality before the change exceeded  $\bar{a}(\rho_1, \tilde{\theta})$ , its quality falls strictly.*

The argument is that a buyer below  $\eta$  must cover the cost of the quality he is served at, and the quality at which he does so is bounded. This is a sufficient condition for a fall, not a characterisation of when falls occur: a seller whose quality already lies below the cap can reach downward without lowering it further, and the proposition says nothing about that case. Section B carries the proof.

A fall in chosen quality need not appear as a fall in measured quality. The measured index responds to the quality shipped and to the volume sold, and reaching downward raises the second while lowering the first — the second of those is a fact about any seller whose access improves, not a possibility. The model therefore predicts a measured decline only where the quality term dominates, and Proposition B.7 in section B states a sufficient condition for that: one buyer below  $\eta$  served after the change, and a certificate  $\mathcal{U}$  that is negative. It is not a characterisation, and Remark B.9 says what it does not cover. Evaluating  $\mathcal{U}$  on a sample is a separate matter and these data do not permit it, because it is built from the cost elasticity, the chosen quality levels before and after, and a revenue-intensity index over the buyer distribution — none of which export records report. The condition is stated because it says what a measured fall requires, not because we evaluate it here.

Two things the estimates already show line up with the mechanism, and neither is a test of it. The mechanism is one a seller carries out on its own: the seller stays, reaches further down and ships something cheaper. That is the margin section 5 locates the decline on, where restricting the estimate to cells present on both sides of the change retains 78% of it, so the model and the estimates are about the same object rather than two objects that happen to point the same way. And reaching down implies a lower price and a larger volume, which is what the continuing-cell estimates of table 7 report in both settings: price

at  $-0.285$  against quantity at  $+0.343$  under permanent normal trade relations, and  $-0.328$  against  $+0.833$  under quota removal. On the full sample the PNTR quantity response is  $-0.098$  instead, so the signature appears on the sample the mechanism is about and not on the one that mixes in the cells that arrive and leave. That last observation is weaker than it looks: restricting to continuing cells removes exits, which are quantity declines, so part of the difference between  $-0.098$  and  $+0.343$  is the restriction rather than the mechanism. The agreement is in direction, and we claim no more from it.

## 6.6 The sufficient index and its regimes

Access, productivity and the size of the market the seller faces do not enter the seller's problem separately. Under the conditions in section A they enter only through their product, a single index  $T$  that scales every variable-profit object the seller weighs against its fixed and relationship costs. A seller with twice the productivity and one with twice the access weight solve the same problem. The model therefore cannot, by itself, tell a market with better access from one with more productive sellers.

What the index buys is a taxonomy. As  $T$  rises, a seller passes through three configurations, with thresholds that are the same for every seller in the market:

- (i) at low  $T$ , the seller's cutoff lies *above*  $\eta$ : it serves only buyers on whom more quality pays, and it offers one variety;
- (ii) at intermediate  $T$ , the cutoff lies *below*  $\eta$  but the seller still offers one variety: it has reached buyers on whom quality does not pay, and Proposition 6.1 applies;
- (iii) at high  $T$ , the seller offers two varieties, serving the buyers below  $\eta$  on the cheaper one.

An improvement in access multiplies  $T$  by the same factor for every seller, so sellers can only move upward through this list. Which transition a given seller makes depends on where it started, and the transitions are what the empirical cells correspond to: a seller that stays in (i) has no reason to lower quality, a seller moving from (i) to (ii) is the case Proposition 6.1 describes, and a seller moving into (iii) is the case section 6.7 describes. The model therefore does not predict that every seller downgrades. It predicts which sellers do, and the answer depends on an object — where a seller's clientele sits relative to  $\eta$  — that the data do not report. Section 7 states what follows from that for the paper's claims.

## 6.7 The menu margin

A seller facing newly reachable low-valuation buyers has two ways to serve them. It can lower the quality of what it already ships, which is the margin above. Or it can keep that variety and add a cheaper one, selling the old variety to the buyers who valued it and the new one to the newcomers.

Which it chooses turns on the fixed cost of the second variety. Opening a variety transfers buyers between varieties rather than adding them, because each buyer takes only one, so what must cover the fixed cost is the gain net of the sales it cannibalises. Writing  $\mathcal{G}(a, \theta^*)$  for that net gain when the existing variety has quality  $a$  and the cutoff is  $\theta^*$ , the seller opens the second variety when

$$\varpi(\rho) \mathcal{G}(a, \theta^*) > F, \quad (9)$$

where  $F$  is the fixed cost of a variety and  $\varpi(\rho)$  is the access weight of section 6.3. Access enters eq. (9) on the left, so improving it makes opening more attractive at a given menu — the same force that widens the clientele also pays for the second variety. And  $\mathcal{G}$  vanishes when the existing variety is already at the bottom of the quality range, because then the second variety is a duplicate of the first and transfers everything while adding nothing. A seller that has already degraded its good will not open a second variety, which is why the two responses are alternatives rather than stages.

This is the observable section 6.1 asked for. A seller that has degraded its good ships one variety, at a lower price, to the same buyers. A seller that has widened its clientele ships two, and the distribution of what it ships gains mass at the bottom while keeping its top. The two accounts differ in the number of varieties and in what happens to the existing one — objects the price, quantity and quality triple does not determine.

The two margins meet the measure in one place, and writing it down says which of them a measured decline can be carrying. A cell that ships varieties  $\ell$  at prices  $p_\ell$  with quantity shares  $s_\ell$  records an average price  $\bar{p} = \sum_\ell s_\ell p_\ell$ , a total quantity  $Q$  and a value  $V = \bar{p}Q$ , so the measured index of eq. (2) is, in the model's objects,

$$\ln \lambda^{\text{meas}} = \frac{\sigma}{\sigma - 1} \ln \bar{p} + \frac{1}{\sigma - 1} \ln Q = \ln \bar{p} + \frac{1}{\sigma - 1} \ln V. \quad (10)$$

Differentiating, with value shares  $\omega_\ell \equiv s_\ell p_\ell / \bar{p}$ ,

$$\Delta \ln \lambda^{\text{meas}} = \underbrace{\sum_\ell \omega_\ell \Delta \ln p_\ell}_{\text{within variety}} + \underbrace{\sum_\ell \frac{p_\ell - \bar{p}}{\bar{p}} \Delta s_\ell}_{\text{between varieties}} + \frac{1}{\sigma - 1} \Delta \ln V, \quad (11)$$

exactly, as a total differential. Degrading the existing variety moves the first term; opening a cheaper one moves the second. A seller with one variety has no second term at all, so the two mechanisms are distinguishable in a cell’s internal composition even though they are not distinguishable in  $\bar{p}$ ,  $Q$  and the residual built from them. Section 5.2 decomposes the aggregate the same way across cells rather than within them; the two are the same arithmetic at different grains.

## 6.8 Varieties and tariff lines

Whether that observable is visible is a fact about the panel rather than about the model, and the answer is: in a minority of cells, carrying a larger minority of value.

Reading the panel this way rests on a mapping we maintain rather than establish. The panel records tariff lines, not varieties, and we identify a model variety with an active HS8 line within a cell. HS8 is an administrative classification and is not a quality ladder, and nothing in the data tells us that a firm’s second line is a lower-quality variety on its menu rather than a different product. Evidence on line entry, line exit or the number of active lines is evidence about reported tariff-line scope; it becomes evidence about a firm’s menu only under this mapping, stated formally as Assumption 7. Every test that reads line structure as menu structure inherits it, and we mark those tests where they appear.

**Table 9:** Lines per cell, and menu transitions across the policy change

| Lines in the cell | Share of cells | Share of value |
|-------------------|----------------|----------------|
| One               | 96.8%          | 86.5%          |
| Two               | 2.9%           | 9.5%           |
| Three or more     | 0.2%           | 4.1%           |
| More than one     | 3.2%           | 13.5%          |

*Notes:* rows are rounded to one decimal and the aggregate row is computed from the unrounded shares, so the parts sum to it only up to that rounding. Cells are firm  $\times$  HS6  $\times$  arm  $\times$  year; a line is an active HS8 code within the cell. Counting only: no regression, no inference, and nothing here is a treatment effect. Reading a line as a menu rung is the maintained mapping stated above.

Two further descriptive facts bear on the exercise. Among units observed in both periods, a little under six percent go from one line to several on the US arm, and those units carry about eleven percent of the arm’s value — so the margin the model points at is not negligible, and it is larger in value than in counts. And where a seller does hold several lines, they are not near-duplicates: the dispersion of measured quality across lines within a cell is a substantial fraction of the dispersion across all cells, which is what the model requires of a menu if the rungs are to differ.

These are counts. They carry no controls, the periods are a calendar split rather than an event, and they rest on the maintained mapping stated at the head of this subsection. They establish that the discriminating margin exists in these data and bound where it can be looked for. They do not test the model, and section 7 says what would.

## 7 Conclusion

Greater access to a market lowers the measured quality of what incumbent Chinese exporters ship to it, and the decline survives inside the same firm shipping the same product to the same arm across the change. Both statements are established for one policy change, the grant of permanent normal trade relations. A second liberalisation of the same market moves the same outcome the same way and by more, which is the kind of agreement a single episode cannot supply. The two part on the margin rather than on the sign: the within-cell result is a PNTR result, and section 4.4 reports the quota estimates with the sensitivity analysis that accompanies them.

That margin is the paper’s contribution. The literature has found quality rising with better access and cheaper inputs (Verhoogen, 2008; Fan, Li, and Yeaple, 2015; Fan, Li, and Yeaple, 2018), and has found it falling through the entry and exit of firms (Khandelwal, Schott, and Wei, 2013; Feng, Li, and Swenson, 2017); the cell in which an incumbent’s own measured quality falls as its access improves has not been filled, and these estimates fill it.

The estimate is narrower than the sentence it compresses into, and the narrowing is worth stating precisely. What the paper estimates is the effect on measured quality of a destination-specific removal of tariff uncertainty, for firm–product–arm cells that ship in both periods, identified from the comparison of a firm’s US and non-US shipments of the same product across products with different exposure. It is not an estimate of the change in the quality of Chinese exports: entry and exit are a different margin, this design does not treat them, and the aggregate that includes them is weighting-dependent, as section 5.2 reports. It is not an estimate of the effect of tariffs, which did not change. And it is one exporting country, one policy change and one treated destination; the mechanism the model describes is general, but the magnitude is not a parameter we have estimated for anywhere else.

What the estimates do not settle is the mechanism, and the paper’s second result is that the standard toolkit cannot settle it either. Measured quality is a linear combination of the price and the quantity it is inferred from, so the three coefficients such a design produces are two independent facts, and no arithmetic on them distinguishes a seller shipping a worse good from a seller shipping the same good to different buyers. This is a property of the measure rather than of our sample, and it applies wherever quality is recovered as a demand

residual. We therefore do not claim that these data prove or distinguish the presence of lower-willingness-to-pay buyers, and no such claim is made anywhere in the paper.

What would settle it is evidence the identity does not determine. The model in section 6 points at one such observable: a seller widening its clientele has a reason to add a cheaper variety and keep its existing one, while a seller degrading its good has none, so the number of varieties and the fate of the existing one separate the two accounts. That margin is visible in these data for a minority of cells carrying a larger minority of value, and reading it as a menu rests on a mapping we maintain rather than identify. The model’s account of a seller reaching below the sorting boundary (section 6.5) describes a subpopulation of sellers selected on an outcome we do not observe — whether a given seller’s clientele crosses that boundary — whose share of sellers we cannot measure; we do not claim that case is typical, and nothing in the paper rests on its being common. Settling the question needs data this panel does not contain: the distribution of transaction prices within a product line, or buyer-level records. Our source is aggregated past both.

We end on the measurement point, because it outlives the application. A large empirical literature reads price, quantity, inferred quality and quality-adjusted price together as evidence about firm behaviour, reporting them on a common right-hand side ([Khandelwal, Schott, and Wei, 2013](#); [Fan, Li, and Yeaple, 2015](#)). Where quality is inferred from the price and the quantity, that reading has one fewer degree of freedom than it appears to, and the mechanism questions it is used to settle are not settled by it. Saying which mechanism is at work requires an observable outside the triple, and the design of future work should start from that rather than discover it afterwards.

## A The Conditions

Section 6 describes the environment in words and states its results under the conditions collected here. This appendix states them. There are twelve: eight that close the model or restrict the region of primitives (section A.1), and four that bind only on the reach or the menu margin, where the statements that name them need them (section A.2). Table 10 records all twelve by block and status.

Of the twelve, the proofs in section B use three — Assumption 2, Assumption 3 and Assumption 5 — together with the two-date configuration of Definition B.1, and section B.1 restates those three in that appendix’s notation. Throughout,  $a \geq 1$  is the quality of a variety and  $\eta > 0$  the elasticity with which quality raises marginal cost,  $\sigma > 1$  is the elasticity of substitution across sellers,  $\theta \in \Theta = [\underline{\theta}, \bar{\theta}]$  is a buyer’s marginal valuation of quality with density  $g > 0$ ,  $k$  is the flow cost of a buyer relationship,  $F_0 a^\kappa$  is the fixed outlay per variety — the  $F$  of eq. (9) — and  $\theta^*$  is the seller’s reach cutoff, with  $\theta_0^*$  and  $\theta_1^*$  its values before and after the access change.

### A.1 The closure and the maintained conditions

Seven of the eight conditions in this block are maintained, and none of them is checkable on the data this paper uses: they are not derived from the environment and this paper does not test them. Assumption 4 is the exception. It is a case hypothesis on one seller’s two cutoffs rather than a maintained restriction, and table 10 classes it accordingly.

**Assumption 1** ((CLOSURE): a scalar price index per source group). Within a date, all buyers face *one* price index for the group of sellers to which the seller belongs:  $P_g(\theta) \equiv P_{g,t}$ , constant across  $\theta$  and moving between the two dates  $\rho_0$  and  $\rho_1$ . The scalar is read at the anchor  $\theta_a = \eta$ .

This is a maintained closure, not a normalisation. The *level* of  $P_g^{\sigma-1}$  is absorbed into the fixed outlay, but the between-date *difference* is not, and it is that difference which does work. The closure is untestable here because no buyer-level price index is observed. Because  $P_g$  does not vary in  $\theta$  under this closure, the type at which the scalar is read is immaterial; the anchor matters only under the  $\theta$ -varying alternative, which this paper does not pursue.

**Assumption 2** ((MON): monotone scale). Write  $\bar{m}(\theta) > 0$  for buyer  $\theta$ ’s spending on the seller’s source group, net of the seller’s own scale — the profile that eq. (B.4) integrates against. Then  $d \ln \bar{m}(\theta) / d\theta > 0$ .

This is what delivers an upper-interval served set (Lemma B.2(i)), and with it the single cutoff  $\theta^*$  that section 6 uses throughout. It is untestable on these data. Under Assumption 1 it is equivalent to the same condition on a buyer's outlay times the share of that outlay which reaches the seller's source group, because a price index that does not vary in  $\theta$  contributes nothing to  $d \ln \bar{m}/d\theta$ .

**Assumption 3** ((NORM): quality normalisation).  $a \geq 1$  for every variety; the admissible quality domain is  $[1, \infty)$ .

This normalises the quality scale and pins the corner at  $a = 1$ . It is the corner at which the net opening gain of eq. (9) vanishes, which is why section 6.7 can say that a seller which has already degraded its good will not open a second variety.

**Assumption 4** ((REGION)).  $\theta_1^* < \eta \leq \theta_0^*$ : the seller's reach cutoff lies at or above the sorting boundary before the access change and strictly below it after.

This is a region restriction, not a testable condition:  $\theta^*$  and  $\eta$  are both unobserved, so it cannot be checked in the data, and this paper offers no test of the region itself. It is a case hypothesis: it binds exactly where a statement names it and is not imposed elsewhere. It names the date pair in which a seller moves from configuration (i) to configuration (ii) of section 6.6. Section 7 records what follows for the paper's claims — a seller in that case is selected on an outcome we do not observe, and we do not claim the case is typical.

**Assumption 5** ((BND): boundedness).  $\kappa > (\sigma - 1)(\bar{\theta} - \eta)$ : the elasticity of the fixed outlay in quality exceeds the largest elasticity of variable profit in quality.

This is an open condition on  $(\kappa, \sigma, \bar{\theta}, \eta)$  and is checkable before any equilibrium is solved. Unlike Assumption 4 it restricts primitives rather than outcomes. Section B uses it only to bound the seller's objective above, so that a maximiser exists; the results proved there use no first-order condition, no second-order condition and no single-crossing or concavity restriction on  $a$  (section B.1).

**Assumption 6** ((SEP): separable varieties). (i)  $g$  is small near  $\eta$ ; and (ii) on a menu of two or more varieties,  $F_0 \leq \bar{F}$ , with a margin delivering a quality separation  $\Delta_a \geq \underline{\Delta} > 0$  between the two varieties. Part (ii) is a condition on a multi-variety menu and has no referent on a one-variety state.

This is a named restriction and it is used wherever a menu carries more than one variety. The bound  $\bar{F}$  is defined by requiring the *endogenous*  $\Delta_a$  to clear  $\underline{\Delta}$ , so the margin relocates the endogeneity rather than removing it. That part (ii) is sufficient for the cross-variety markup terms to vanish is a conjecture and not a result of this paper.

**Assumption 7** ((LINE): the line–variety mapping). A model variety is identified with an active HS8 line within a cell.

A maintained mapping, not a conclusion identified from data. Section 6.8 states it where the model uses it and marks the tests that inherit it: HS8 is an administrative classification, and nothing in the data says that a seller’s second line is a cheaper variety of its menu rather than a different product. The prohibition attached to it is at section 7 — the paper does not claim that its results show the data prove or distinguish buyers by willingness to pay.

**Assumption 8** ((ARM): common quality across destination arms). A variety is a *product* rather than a destination-specific specification, so one quality  $a$  serves both destination arms.

This is substantive and its scope is narrow. No result in section 6 or section B requires it; it is needed only by statements that read the model across both arms of section 3’s comparison. Where a statement uses it, it says so, and the results that do not use it are not weakened by it.

## A.2 The margin conditions

*Remark A.1* (No separate participation cutoff in the reverted state). The model carries no seller-side participation cutoff and no separate buyer-side cutoff in the reverted tariff state. Participation is  $\varpi(\rho)v(\theta) \geq k$ , with  $v(\theta)$  the good-state value of a relationship with buyer  $\theta$  and  $\varpi(\rho)$  the access weight of section 6.3. The reverted state is the same environment at a different tariff, which forces the reverted-state value to be  $\chi v(\theta)$  for a scalar  $\chi \in (0, 1)$  varying neither in  $\theta$  nor in  $a$ . The flow cost  $k$  is paid in both states, so there is no seller-side cutoff; and in the fixed-schedule reading — clause (i) of Definition B.1 — the buyer-side schedules do not move, so there is no buyer-side cutoff either. The reverted-state served set therefore equals the good-state served set, and the upper-interval conclusion of Lemma B.2(i) follows from Assumption 2 alone, with no jump. Outside the fixed-schedule reading a reverted-state schedule exists and a second cutoff can re-emerge.

**Assumption 9** ((X-L): the gross condition for the cheaper variety). Write  $\Pi_v(a; S)$  for the seller’s variable profit from clientele  $S$  at quality  $a$ , and let  $\iota(\theta^*) \equiv \Pi_v(1; [\theta^*, \eta])$  be the *gross* index of the cheaper variety: what it would earn on the buyers below the sorting boundary, at the quality corner. Weighed against the fixed outlay,  $\iota$  is **necessary for the cheaper variety to be open and is not sufficient**. What decides openness is eq. (9), with the net gain there given by

$$\mathcal{G}(a, \theta^*) \equiv \Pi_v(1; [\theta^*, \eta]) - \Pi_v(a; [\theta^*, \eta]). \quad (\text{A.1})$$

Opening the cheaper variety *transfers* the buyers in  $[\theta^*, \eta)$  from the existing variety rather than *adding* them, so what must cover the fixed outlay is the gain net of cannibalisation. The transfer term vanishes exactly at  $a = 1$ , where  $\mathcal{G}(1, \theta^*) \equiv 0$ ; hence  $\iota \geq F_0$  does not make the deviation pay. One consequence is worth recording. A statement that there is at most one access threshold  $\bar{\rho}$  at which the cheaper variety opens is a statement about the *gross* index  $\iota$ , and it does not transfer to the locus at which the cheaper variety actually opens. The uniqueness of that locus is not established.

**Assumption 10** ((A-FLOOR): a quality floor). There is  $\underline{a} > 1$  with  $a(\rho) \geq \underline{a}$  for every  $\rho \in [\rho_0, \rho_1]$ .

This is a genuine restriction and not a free strengthening.  $\mathcal{G}$  depends on *two* states, the existing variety's quality  $a$  and the cutoff  $\theta^*$ , and a condition that stops the cutoff from jumping upward controls only one of them. Nothing in the model stops  $a$  from falling to the quality corner, where  $\mathcal{G}(1, \theta^*) \equiv 0$  by eq. (A.1) and the cheaper variety closes at any level of the gross index. Assumption 10 and the strict form of eq. (9) are a joint hypothesis on one  $\underline{a}$  and are not separable.

**Assumption 11** ((REG- $\Lambda$ ): regular loop gain along the path). Write  $\Lambda$  for the loop gain of the joint reach–quality system: the product of the response of the cutoff to quality and of quality to the cutoff, so that a direct effect becomes a total effect through the multiplier  $1/(1 - \Lambda)$ . A  $\rho$ -path satisfies (REG- $\Lambda$ ) if no configuration on it has  $\Lambda = 1$ .

Assumed wherever it is invoked and proved nowhere. At  $\Lambda = 1$  the multiplier is undefined and the derivative along the path need not exist. The condition is not an optimality requirement:  $\Lambda > 1$  is excluded because it is a saddle, but  $\Lambda = 1$  is excluded only by assumption. It is the price of differentiating the joint system of section 6.4, and section B does not pay it, because the cap it proves is a feasibility restriction rather than a first-order condition.

**Assumption 12** ((CUT-MON): cutoff monotonicity).  $\theta_i^*$  weakly falls between the two dates for every seller  $i$ .

Stated, not proved. Nothing in the model delivers it: reach and quality are solved jointly (section 6.4), and a rise in the access index moves both. Like Assumption 4, and unlike Assumption 5, it restricts outcomes rather than primitives.

**Table 10:** The twelve conditions, by block and status

| Tag       | Condition     | Block      | Status                                                                                               |
|-----------|---------------|------------|------------------------------------------------------------------------------------------------------|
| (CLOSURE) | Assumption 1  | Maintained | Maintained closure; untestable here                                                                  |
| (MON)     | Assumption 2  | Maintained | Maintained; untestable here                                                                          |
| (NORM)    | Assumption 3  | Maintained | Normalisation of the quality scale                                                                   |
| (REGION)  | Assumption 4  | Case       | Case hypothesis on one seller’s two cutoffs; binds where named; cannot be checked in the data        |
| (BND)     | Assumption 5  | Maintained | Checkable in the parameters; used only for existence                                                 |
| (SEP)     | Assumption 6  | Maintained | Named restriction; part (ii) binds only on a multi-variety menu, and its sufficiency is a conjecture |
| (LINE)    | Assumption 7  | Maintained | Maintained mapping; not a conclusion identified from data                                            |
| (ARM)     | Assumption 8  | Maintained | Substantive; needed only by cross-arm statements                                                     |
| (X-L)     | Assumption 9  | Margin     | Necessary, not sufficient; eq. (9) decides openness                                                  |
| (A-FLOOR) | Assumption 10 | Margin     | Assumed; proved nowhere                                                                              |
| (REG-A)   | Assumption 11 | Margin     | Assumed; proved nowhere                                                                              |
| (CUT-MON) | Assumption 12 | Margin     | Assumed; proved nowhere                                                                              |

*Notes:* “Maintained” marks a condition the paper imposes and does not test. “Case” marks a hypothesis that binds only in statements that name it. “Margin” marks a condition invoked only on the reach or the menu margin. Remark A.1 is a remark on the model rather than a condition and is not counted here.

## B The reach cap, and when it is visible in measured quality

This appendix proves Proposition 6.1 and states the condition the main text refers to when it says that a fall in chosen quality need not appear as a fall in measured quality. Section B.1 fixes the seller's problem in the notation of section 6; section B.2 is the cap; section B.3 is the mapping into the measured index.

### B.1 The seller's problem

A seller of productivity type  $x \equiv \varphi^{\sigma-1}$  chooses a quality  $a \geq 1$  and a served set of buyers. Marginal cost is  $c_0 a^\eta / \varphi$ , so quality raises marginal cost with elasticity  $\eta > 0$ , and constant-elasticity demand with elasticity  $\sigma > 1$  gives the constant-markup price

$$p(a) = \frac{\sigma}{\sigma-1} \frac{c_0}{\varphi} a^\eta, \quad \text{so} \quad p(a) a^{-\eta} \text{ does not depend on } a. \quad (\text{B.1})$$

Buyer  $\theta$  spends  $m(\theta)$  at a quality-adjusted price, a relationship with him costs the flow  $k$ , and a variety costs the fixed outlay  $F_0 a^\kappa$ . Writing  $\bar{m}$  for the type profile of spending net of the seller-specific scale,  $g$  for the density of types on  $\Theta = [\underline{\theta}, \bar{\theta}]$ , and

$$T \equiv c \varpi(\rho) x \quad (\text{B.2})$$

for the access index — the seller's own scale times the weight  $\varpi(\rho)$  that section 6.3 places on variable profit, strictly increasing in  $\rho$  — the value of the one-variety configuration at a served set  $[\theta^*, \bar{\theta}]$  is

$$\pi(a, \theta^*; T) = T \bar{\Gamma}(a; [\theta^*, \bar{\theta}]) - F_0 a^\kappa - k \mathbf{N}(\theta^*), \quad (\text{B.3})$$

where

$$\bar{\Gamma}(a; S) \equiv \int_S \bar{c} \bar{m}(\theta) a^{(\sigma-1)(\theta-\eta)} g(\theta) d\theta, \quad \mathbf{N}(\theta^*) \equiv \int_{\theta^*}^{\bar{\theta}} g(\theta) d\theta. \quad (\text{B.4})$$

The relationship cost and the fixed outlay are undiscounted, so (B.3) is *affine in T*: the whole of the policy enters as a scalar weight on the first term, which is the sentence section 6.3 makes in words. Realised revenue at the same state is

$$V = \sigma c x \tau \bar{\Gamma}(a; [\theta^*, \bar{\theta}]), \quad (\text{B.5})$$

with  $\tau$  collecting the date- and destination-specific factors that are outside the seller's choice. Note that  $\varpi$  does not appear in (B.5): it weights the *expected* value of a relationship across the two tariff states, not the revenue recorded in the good state.

**Conditions invoked.** The proofs below use three of the conditions collected in section A, restated here in this section's notation, and nothing else.

- (a) **(MON)**  $d \ln \bar{m}(\theta)/d\theta > 0$  at almost every  $\theta$ : spending rises with the taste for quality.
- (b) **(NORM)**  $a \geq 1$ : the admissible quality domain is  $[1, \infty)$ .
- (c) **(BND)**  $\kappa > (\sigma - 1)(\bar{\theta} - \eta)$ : the fixed outlay outgrows the revenue gain from quality, so (B.3) is bounded above and a maximiser exists.

**(BND)** is used only for existence. Neither the cap nor its consequence uses a first-order condition, a second-order condition, or any single-crossing or concavity restriction on  $a$ ; that is what makes them survive configurations in which first-order conditions do not locate the optimum.

**Definition B.1** ((CONF): the configuration the results are stated on). A seller's pair of dates satisfies (CONF) if: (i) the buyer-side schedules  $(P, \psi)$  are the same at the two dates, so that  $\bar{m}$  and  $g$  do not move; (ii) exactly one variety is active at each date; (iii) the factors collected in  $\tau$  are common to the two dates; and (iv) the access change raises the index,  $T_1 > T_0$ .

(CONF) is not a new condition on primitives. It names the configuration — fixed schedules, one active variety, no change in the outside factors — on which the two-date statements below are made, and table 9 is what says how much of the panel sits in clause (ii). Where a result needs (CONF), it says so.

Differentiating (B.4) under the integral, on any  $S$  with  $\bar{m}g$  integrable,

$$T \frac{\partial \bar{\Gamma}(a; S)}{\partial \ln a} = \int_S (\theta - \eta) \omega_a(\theta) d\theta, \quad \omega_a(\theta) \equiv (\sigma - 1) T \bar{c} \bar{m}(\theta) a^{(\sigma-1)(\theta-\eta)} g(\theta) > 0, \quad (\text{B.6})$$

which is eq. (7): the marginal value of quality to a seller serving  $S$  is the integral of  $(\theta - \eta)$  against a positive weight proportional to what each served buyer spends. Buyers above  $\eta$  contribute positively and buyers below it negatively, and the sign of the integrand flips exactly at the cost elasticity. Everything below is a consequence of that display and of the participation margin.

## B.2 The reach cap

Define the *participation index* of buyer  $\theta$  at quality  $a$  and access index  $T$ ,

$$\iota(\theta; a, T) \equiv T \bar{c} \bar{m}(\theta) a^{(\sigma-1)(\theta-\eta)}, \quad (\text{B.7})$$

so that from (B.3) and (B.4),

$$\frac{\partial \pi}{\partial \theta^*} = g(\theta^*) [k - \iota(\theta^*; a, T)]. \quad (\text{B.8})$$

**Lemma B.2** (Served buyers clear the flow cost). *Assume (MON) and (NORM). Then*

- (i)  $\theta \mapsto \iota(\theta; a, T)$  is strictly increasing on  $\Theta$ , so the set  $\{\theta : \iota(\theta; a, T) \geq k\}$  is an upper interval and the seller's served set is  $[\theta^*, \bar{\theta}]$  for some  $\theta^*$ ;
- (ii) at any maximiser  $(a, \theta^*)$  of (B.3),  $\iota(\theta^*; a, T) \geq k$ , with equality whenever  $\theta^* > \underline{\theta}$ ; consequently  $\iota(\theta; a, T) \geq k$  for every served  $\theta$ .

*Proof. Step 1 (part (i)).* From (B.7),  $d \ln \iota / d\theta = d \ln \bar{m}(\theta) / d\theta + (\sigma - 1) \ln a$ . The first term is strictly positive by (MON) and the second is non-negative by (NORM), so  $\iota$  is strictly increasing in  $\theta$ . A strictly increasing function exceeds a threshold on an upper interval, and the seller serves  $\theta$  exactly when the relationship pays its flow cost, i.e. when  $\iota(\theta; a, T) \geq k$ .

*Step 2 (part (ii), interior cutoff).* If  $\theta^* > \underline{\theta}$  is optimal then  $\partial \pi / \partial \theta^* = 0$  there, and by (B.8) with  $g > 0$  this is  $\iota(\theta^*; a, T) = k$ .

*Step 3 (part (ii), corner cutoff).* Suppose  $\theta^* = \underline{\theta}$  and  $\iota(\underline{\theta}; a, T) < k$ . Then (B.8) gives  $\partial \pi / \partial \theta^* = g(\underline{\theta})[k - \iota(\underline{\theta}; a, T)] > 0$ , so raising  $\theta^*$  above  $\underline{\theta}$  — dropping the marginal buyers — strictly raises  $\pi$ , contradicting optimality. Hence  $\iota(\underline{\theta}; a, T) \geq k$ . Note that this step compares values and uses no derivative in  $a$ .

*Step 4.* By Steps 2–3 the index is at least  $k$  at the cutoff; by Step 1 it is increasing in  $\theta$ ; so it is at least  $k$  at every  $\theta \geq \theta^*$ .  $\square$

**Lemma B.3** (The reach cap). *Assume (MON) and (NORM), and let  $(a, \theta^*)$  be any maximiser of (B.3) at access index  $T$ . If the maximiser serves a buyer  $\tilde{\theta} < \eta$ , then*

$$a \leq \bar{a}(T, \tilde{\theta}) \equiv \left( \frac{T \bar{c} \bar{m}(\tilde{\theta})}{k} \right)^{\frac{1}{(\sigma-1)(\eta-\tilde{\theta})}}, \quad (\text{B.9})$$

and  $\bar{a}(T, \tilde{\theta}) \geq 1$ . Moreover, writing  $d \equiv \eta - \tilde{\theta} > 0$  for the distance of the buyer below the cost elasticity,  $\bar{a}$  is

- (i) strictly increasing in  $T$ , hence strictly increasing in  $\rho$  along (B.2); and

(ii) *strictly decreasing in  $d$  on the range of  $d$  at which  $\tilde{\theta}$  can be served at all.*

*Proof. Step 1 (the cap).* By Lemma B.2(ii) the served buyer  $\tilde{\theta}$  satisfies  $T\bar{c}\bar{m}(\tilde{\theta})a^{(\sigma-1)(\tilde{\theta}-\eta)} \geq k$ , i.e.

$$a^{-(\sigma-1)(\eta-\tilde{\theta})} \geq \frac{k}{T\bar{c}\bar{m}(\tilde{\theta})}. \quad (\text{B.10})$$

Because  $\tilde{\theta} < \eta$  the exponent on the left is strictly negative, so raising both sides to the power  $-1/[(\sigma-1)(\eta-\tilde{\theta})] < 0$  reverses the inequality and gives (B.9).

*Step 2 ( $\bar{a} \geq 1$ , and what it means).* By (NORM),  $a \geq 1$ , so the left side of (B.10) is at most 1; hence  $T\bar{c}\bar{m}(\tilde{\theta}) \geq k$ , and  $\bar{a}(T, \tilde{\theta}) \geq 1$ . So the cap is never vacuous in the wrong direction: a buyer below  $\eta$  whom no admissible quality can make worth serving is one for which  $T\bar{c}\bar{m}(\tilde{\theta}) < k$ , and such a buyer is simply not served.

*Step 3 (monotonicity in  $T$ ).*  $\ln \bar{a} = \ln(T\bar{c}\bar{m}(\tilde{\theta})/k)/[(\sigma-1)d]$  is strictly increasing in  $T$  at fixed  $d$ , because the numerator is and the denominator is positive and constant.  $T$  is strictly increasing in  $\rho$  by (B.2) and the monotonicity of  $\varpi$ .

*Step 4 (monotonicity in  $d$ ).* Write  $N(d) \equiv \ln(T\bar{c}\bar{m}(\eta-d)/k)$ , so that  $\ln \bar{a} = N(d)/[(\sigma-1)d]$ . By Step 2, a buyer who is served at all has  $N(d) \geq 0$ ; by (MON),  $N$  is strictly decreasing in  $d$  wherever  $\bar{m}$  is strictly increasing, and non-increasing everywhere. The factor  $1/[(\sigma-1)d]$  is strictly positive and strictly decreasing in  $d$ . A non-negative non-increasing function times a positive strictly decreasing function is strictly decreasing, so  $\ln \bar{a}$ , and hence  $\bar{a}$ , is strictly decreasing in  $d$ .  $\square$

Two readings of Lemma B.3 are worth separating. It is a *feasibility restriction on the optimum itself*, not a comparison of slopes: it says that serving a buyer who dislikes quality is incompatible with high quality, whatever the profit function looks like in between. And it is the model's statement of the sentence in section 6.2: the low-valuation buyers that better access makes reachable are reachable *only* at low quality. The seller cannot have both.

*Proof of Proposition 6.1. Step 1 (the cap).* The proposition's first sentence is Lemma B.3, with  $\bar{a}(\rho, \tilde{\theta}) \equiv \bar{a}(T(\rho), \tilde{\theta})$  evaluated at the seller's own index (B.2); the monotonicity of  $\bar{a}$  in the distance of  $\tilde{\theta}$  below  $\eta$  is Lemma B.3(ii).

*Step 2 (the strict fall).* Let the seller's date- $j$  optimum be  $(a_j, \theta_j^*)$  at index  $T_j$ ,  $j = 0, 1$ , and suppose the date-1 optimum serves some  $\tilde{\theta} < \eta$ . By Lemma B.3 applied at date 1,

$$a_1 \leq \bar{a}(\rho_1, \tilde{\theta}). \quad (\text{B.11})$$

If in addition  $a_0 > \bar{a}(\rho_1, \tilde{\theta})$ , then  $a_1 \leq \bar{a}(\rho_1, \tilde{\theta}) < a_0$ , so quality falls strictly.  $\square$

*Remark B.4* (What Proposition 6.1 does and does not say). Three restrictions travel with it, and the main text states all three.

- (i) **It is sufficient for a strict fall, not a characterisation.** A seller whose quality already lies weakly below  $\bar{a}(\rho_1, \tilde{\theta})$  can extend its reach below  $\eta$  without lowering quality at all, and Proposition 6.1 says nothing about that case. Nothing below converts the cap into an “only if”.
- (ii) **The cap that binds is the post-change one.**  $\bar{a}$  rises with access (Lemma B.3(i)), so the comparison in Step 2 of the proof is  $a_0$  against  $\bar{a}(\rho_1, \tilde{\theta})$ , not against  $\bar{a}(\rho_0, \tilde{\theta})$ . Using the pre-change cap would be the weaker and wrong test.
- (iii) **The cap has content only for a specific low  $\tilde{\theta}$ .**  $\bar{a}(\rho, \tilde{\theta}) \rightarrow \infty$  as  $\tilde{\theta} \uparrow \eta$ , so knowing only that the date-1 cutoff lies below  $\eta$  delivers no bound. The hypothesis is that a *named* buyer strictly below  $\eta$  is served at date 1.

### B.3 From chosen quality to measured quality

The main text says that a fall in chosen quality need not appear as a fall in the measured index. This subsection says why, and states the condition under which it does.

**Lemma B.5** (The measured index splits into a quality term and a revenue-intensity term). *Under (CONF), with  $\lambda$  the measured quality index of eq. (2),*

$$\Delta \ln \lambda = \underbrace{\eta \Delta \ln a}_{\text{quality term}} + \underbrace{\frac{1}{\sigma-1} \Delta \ln \bar{\Gamma}^*}_{\text{revenue-intensity term}}, \quad \bar{\Gamma}_j^* \equiv \bar{\Gamma}(a_j; [\theta_j^*, \bar{\theta}]). \quad (\text{B.12})$$

*Proof. Step 1.* Equation (2) is  $\ln \lambda = \frac{\sigma}{\sigma-1} \ln p + \frac{1}{\sigma-1} \ln q$ , which rearranges to  $\ln \lambda = \ln p + \frac{1}{\sigma-1} \ln(pq) = \ln p + \frac{1}{\sigma-1} \ln V$ . With one active variety at each date ((CONF)(ii)) the cell's price is that variety's price and its revenue is that variety's revenue, so no share-weighting convention is needed.

*Step 2.* By (B.1),  $\ln p = \eta \ln a + \text{const}$ , where the constant depends on  $(\sigma, c_0, \varphi)$  and not on  $a$  or on the served set; by (CONF)(i) it does not move between the dates. By (B.5),  $\ln V = \ln \bar{\Gamma}^* + \ln(\sigma c x \tau)$ , and the second term does not move between the dates by (CONF)(i) and (iii).

*Step 3.* Substituting Step 2 into Step 1 and differencing, the constants cancel and (B.12) follows.  $\square$

**Lemma B.6** (The revenue-intensity term never falls). *Under (CONF),  $\bar{\Gamma}_1^* \geq \bar{\Gamma}_0^*$  for any selection of maximisers at the two dates. Consequently measured quality falls only if chosen quality falls.*

*Proof. Step 1.* Write  $z_j = (a_j, \theta_j^*)$ ,  $\bar{\Gamma}(z)$  for (B.4) at  $z$ , and  $K(z) \equiv F_0 a^\kappa + kN(\theta^*)$ , so that (B.3) reads  $\pi(z; T) = T\bar{\Gamma}(z) - K(z)$ , affine in  $T$  with slope  $\bar{\Gamma}(z) \geq 0$ . Neither  $\bar{\Gamma}$  nor  $K$  depends on  $T$ .

*Step 2.* Optimality of  $z_0$  at  $T_0$  and of  $z_1$  at  $T_1$  gives

$$T_0\bar{\Gamma}(z_0) - K(z_0) \geq T_0\bar{\Gamma}(z_1) - K(z_1), \quad T_1\bar{\Gamma}(z_1) - K(z_1) \geq T_1\bar{\Gamma}(z_0) - K(z_0).$$

Adding and cancelling the  $K$  terms,  $(T_1 - T_0)[\bar{\Gamma}(z_1) - \bar{\Gamma}(z_0)] \geq 0$ , and  $T_1 > T_0$  by (CONF)(iv), so  $\bar{\Gamma}(z_1) \geq \bar{\Gamma}(z_0)$ .

*Step 3.* By Lemma B.5 the second term of (B.12) is then non-negative, so  $\Delta \ln \lambda < 0$  requires  $\eta \Delta \ln a < 0$ , and  $\eta > 0$  gives  $\Delta \ln a < 0$ .  $\square$

Lemma B.6 is the formal content of the main text's warning. The two terms of (B.12) move in opposite directions when a seller reaches down: the quality term falls by Proposition 6.1, and the revenue-intensity term weakly rises, because a seller that has just been given better access does not end up selling less. Whether the measured index falls is therefore a question about magnitudes, and the following proposition is the condition the main text points at.

**Proposition B.7** (When the quality term dominates). *Assume (MON), (NORM), (BND) and (CONF). Suppose the date-1 optimum serves a buyer  $\tilde{\theta} < \eta$ , write  $\bar{a} \equiv \bar{a}(\rho_1, \tilde{\theta})$  for the cap of (B.9) at date 1, and let  $\bar{\Gamma}_\Theta(a) \equiv \bar{\Gamma}(a; \Theta)$  be the revenue-intensity index of the widest possible clientele. Define*

$$\mathcal{U} \equiv \eta [\ln \bar{a} - \ln a_0] + \frac{1}{\sigma - 1} \ln \frac{\max_{1 \leq a \leq \bar{a}} \bar{\Gamma}_\Theta(a)}{\bar{\Gamma}_0^*}. \quad (\text{B.13})$$

*Then  $\Delta \ln \lambda \leq \mathcal{U}$ . In particular, if  $\mathcal{U} < 0$  then measured quality falls.*

*Proof. Step 1 (the quality term).* Lemma B.3 at date 1 gives  $a_1 \leq \bar{a}$ . Since  $\eta > 0$ ,  $\eta \Delta \ln a = \eta [\ln a_1 - \ln a_0] \leq \eta [\ln \bar{a} - \ln a_0]$ .

*Step 2 (the revenue-intensity term).* The integrand of (B.4) is positive, so enlarging the clientele enlarges the integral:  $\bar{\Gamma}_1^* = \bar{\Gamma}(a_1; [\theta_1^*, \tilde{\theta}]) \leq \bar{\Gamma}_\Theta(a_1)$ . By (NORM) and Step 1,  $a_1 \in [1, \bar{a}]$ , so  $\bar{\Gamma}_\Theta(a_1) \leq \max_{1 \leq a \leq \bar{a}} \bar{\Gamma}_\Theta(a)$ . The maximum is attained because  $\bar{\Gamma}_\Theta$  is continuous on the compact interval  $[1, \bar{a}]$ .

*Step 3.* Substituting Steps 1 and 2 into Lemma B.5's (B.12) gives  $\Delta \ln \lambda \leq \mathcal{U}$ . If  $\mathcal{U} < 0$  then  $\Delta \ln \lambda < 0$ .  $\square$

*Remark B.8* (The maximum in (B.13) is not at the endpoint). The second term of (B.13) must be evaluated as a maximum over  $[1, \bar{a}]$  and *not* at  $a = \bar{a}$ . By (B.6) applied on  $S = \Theta$ ,

$$\frac{d\bar{\Gamma}_\Theta(a)}{d \ln a} = (\sigma - 1) \int_\Theta (\theta - \eta) \bar{c} \bar{m}(\theta) a^{(\sigma-1)(\theta-\eta)} g(\theta) d\theta,$$

whose sign is the sign of the spending-weighted average of  $(\theta - \eta)$  over the *whole* type support. Where that support carries enough mass below  $\eta$  the derivative is negative throughout  $[1, \bar{a}]$ ,  $\bar{\Gamma}_\Theta$  is strictly decreasing there, and the maximum is at  $a = 1$ , namely  $\bar{\Gamma}_\Theta(1) = \int_\Theta \bar{c} \bar{m} g$ . In that case  $\bar{\Gamma}_\Theta(\bar{a})$  is the *minimum* on the interval, and substituting it into (B.13) would understate  $\mathcal{U}$  and would not bound  $\Delta \ln \lambda$  at all. The certificate is a bound only when the maximum is computed; it is one line from the primitives, and the sign of the derivative above says which endpoint to check.

*Remark B.9* (Scope of Proposition B.7). (i) **Sufficient, not necessary.** A seller with  $\mathcal{U} > 0$  can still record a fall. The bound discards two things — the date-1 optimum's actual quality and its actual clientele — and both discards push in the same direction.

- (ii) **It needs one below- $\eta$  buyer *known* to be served at date 1.** What Proposition B.7 removes is the first-order comparison between the two dates, not the identification of whom the seller reaches: knowing only that the date-1 cutoff lies below  $\eta$  is not enough, because the cap is unbounded as  $\tilde{\theta} \uparrow \eta$  (Remark B.4(iii)). The inputs are the primitives, the date-0 optimum, and one named below- $\eta$  type served at date 1.
- (iii) **It is a statement about one seller, in the configuration (CONF).** It says nothing about a seller that opens a second variety, which is the other margin of section 6.7, and it aggregates to nothing without a further argument about the distribution of sellers.

## C The input-tariff channel, on a ladder of fixed effects

This appendix reports what the input-tariff lever does to measured quality and to price once the firm controls that absorb it are taken off in steps. It supports section 4.2, where the ladder table and the estimates are, and it carries what that section states without developing: how the lever is built, what the price column does, what separates the top and bottom rungs, and the separability test whose result is quoted there.

## C.1 The lever, and why a ladder is needed

The input lever is a firm’s exposure to China’s tariff cuts on imported inputs. We measure it as the firm’s 2001 ordinary-import basket, valued at contemporaneous applied rates and downweighted by the share of that firm’s 2001 imports entering under the statutorily duty-exempt processing regime. It is a firm–year object: one number per firm per year, the same for every product that firm ships and every destination it ships to. On the estimation sample it falls from a cell-weighted mean of 9.86 percentage points in 2000 to 2.81 in 2011, and its standard deviation is 5.55 percentage points.

**That object has no within-firm–year variation at all, and the absorption section 4.2 describes is exact rather than approximate.** Of the 196,288 firm–year cells in the PNTR estimation sample, none carries more than one value of the lever. Under the fixed effects of eq. (3) the lever is perfectly collinear with the firm  $\times$  year effects and is dropped, with and without the access terms in the regression. No coefficient on it exists in the specification this paper estimates, and none can be made to exist without removing the control that protects the access design.

Comparing the two levers therefore requires a ladder. We estimate the input effect at four rungs, which differ only in how much firm-level heterogeneity they hold fixed:

- **L0** — firm  $\times$  year, HS6  $\times$  year and arm  $\times$  year effects. This is eq. (3), whose third effect has two groups in each year rather than one per destination (section 3.1). The lever is absorbed.
- **L1** — firm, HS6  $\times$  year and arm  $\times$  year effects. Permanent firm heterogeneity is held fixed; the lever is identified off cross-firm differences in exposure over time. This is the cleanest rung at which the input effect exists.
- **L2** — firm  $\times$  HS6 and year effects. Firm–product heterogeneity is held fixed, the product–year and arm–year controls are not.
- **L3** — HS6 and year effects. No firm control of any kind.

The lever falls with liberalisation, so we report every estimate in the liberalisation direction, as the percentage change in the outcome per 10 percentage point cut in the firm’s input tariff. A positive number is the direction Fan, Li, and Yeaple (2015) and Bas and Strauss-Kahn (2015) report: cheaper inputs, higher measured quality and higher price. The  $t$ -statistics are reported with the same orientation, so that sign and  $t$  agree throughout.

## C.2 The ladder

Table 5 is in section 4.2, together with the quality estimates, the rule for reading them and the comparison against the published magnitudes. What follows is what that section does

not have room for: what the price column does, what the gap between the top and bottom rungs measures, and the direction in which the two levers were tested for separability.

### C.3 The price column, and the rung at which it reverses

**At L2 the price estimate is significant and carries the wrong sign for the input-tariff literature.** A 10 percentage point cut in the input tariff is associated with a price that is 2.49% *lower*, with an interval of  $[-3.82, -1.17]$  and  $t = -3.70$  on the firm-product sample of 5,546,425 rows. Cheaper inputs raising export prices is the input channel's prediction; this rung delivers the opposite, and delivers it with a confidence interval that excludes zero. The effect is small — about 1.6% of the within-effect standard deviation of price at that rung — but it is not zero and it is not the sign the channel predicts.

**It is the one genuine significant-to-significant sign reversal in table 5**, running from L2's  $-2.49\%$  to L3's  $+7.78\%$ , and it is on the outcome the input-tariff literature actually measures. Our claim in section 4.2 is that the input channel does not generate the access result, and a wrong-signed input effect on price does not rescue that claim — it simply fails to supply the positive one the alternative reading needs.

### C.4 What the gap between L3 and L1 measures

The two rungs differ in one thing only: L1 holds permanent firm heterogeneity fixed and L3 does not. Whatever separates their estimates is therefore what the firm control was blocking. **On quality that separation is about 16 percentage points per 10 percentage point cut, and on price about 9:  $+15.40\%$  against  $-0.53\%$ , and  $+7.78\%$  against  $-1.04\%$ .**

We offer the following as an interpretation of that gap and not as an estimate of anything. The cross-firm variation L3 adds back is the variation that is already known to be endogenous here: firms on rising quality trajectories were also the firms whose 2001 import baskets faced the deeper cuts. Without a firm control, a firm that was upgrading for its own reasons and a firm that faced a larger input cut cannot be told apart, and the upgrading is booked to the tariff. On that reading essentially none of L3's magnitude is the input cut's doing, and the size of the gap is the size of the confound. Nothing in the ladder tests this reading; it is the reading that makes the ladder's pattern coherent, and a reader is free to prefer another.

One observation bears on which rung is biased. **If the firm effects were overcontrolling a real and large input effect, removing them in steps should move the estimate toward that effect with progressively overlapping intervals.** That is not what table 5 shows. The estimate sits at zero through two rungs and then jumps, at the rung where the last firm control comes off, to a value incompatible with the intervals of the

rungs before it. Sharpening is what an unmasked effect looks like; a jump at the final rung is what a confound looks like.

The L3 price estimate of +7.78% lands inside FLY’s reported band of +5.6% to +8.2% per 10 percentage point cut. **The literature’s magnitude is reproducible on these data, and the only specification that reproduces it is the one with no firm control at all.** This does not refute FLY. Their design identifies the input effect within the firm across output products, using an input–output weighting our firm-basket lever does not carry, and that design cannot be run on this panel as the lever is currently built. What the ladder shows is what this lever gives at each rung, and where on the ladder the published magnitude appears.

## C.5 Separability, in the direction we measured it

Every rung was estimated twice, once with the input lever alone and once with the access triple difference and its lower-order terms added as controls. Table 11 reports how much the input coefficient moves when the access block is added. **It moves by at most 5.2% of its own magnitude, at every rung and on both outcomes.** The two levers are separable in this direction, which is what their variation predicts: the input lever is a firm–year object and the access lever varies across products and arms within the firm–year.

**Table 11:** Movement in the input coefficient when the access controls are added

| Rung | Measured quality $\ln \lambda$ | Price $\ln p$ |
|------|--------------------------------|---------------|
| L1   | 3.5%                           | 1.4%          |
| L2   | 0.8%                           | 5.2%          |
| L3   | 3.1%                           | 4.8%          |

*Notes:* each entry is the absolute change in the input coefficient when the access triple difference and its lower-order terms are added to the specification, as a percentage of the input coefficient’s own magnitude in the specification without them. Samples and fixed effects are those of table 5. L0 is omitted: the input lever is absorbed there with and without the access controls, so there is no coefficient to move. The table reports the movement of the input coefficient only; see section C.5 on what it does not report.

**The converse has not been run.** The exercise behind table 11 retains only the input terms in its output. How the access coefficient of eq. (3) moves when the firm’s input-tariff exposure is added as a control is not reported by it, and no table in this paper reports it. **We therefore make no claim that controlling for input exposure leaves the access coefficient unchanged.** The separability this appendix establishes runs in one direction: adding an access control barely moves the input coefficient. That direction has a build behind it and the other does not.

There is a reason the missing direction is harder than it sounds, and it is the same reason this appendix exists. Under the fixed effects of eq. (3) the input lever is absorbed exactly, so it cannot be added to the baseline specification as a control at all — there is nothing to add. Running the converse means either weakening the firm  $\times$  year effects, which is the identification the access design rests on, or rebuilding the input lever so that it varies within the firm-year across output products. Neither has been done here, and the claim that the access estimate survives an input-exposure control is not one this paper makes.

## D Sample Construction and the Treatment Arm

This appendix documents what section 2 states in a sentence each: how the estimation sample is built from the customs records, the transaction rules behind the cell price and quantity, where the exposure measure comes from and how it is carried across tariff revisions, what the treated arm resolves to, how the inputs to measured quality are winsorised, and what restricting the arm to the fifty states would change.

### D.1 From customs records to estimation cells

The source is a transaction ledger. It carries 149,357,863 records for 2000 through 2014, of which 105,490,292 are exports; imports are dropped first and play no further part. Table 12 reports the count surviving each subsequent step.

Two features of the flow are not carried by the labels.

**Almost nothing is dropped for an unreadable product code, and nothing at all is dropped for its destination.** Fifty-two export records in fifteen years carry a tariff code that does not resolve to a six-digit product, and they are removed; they are worth USD 622,844 against an export ledger of USD 17,535 billion. Every export record in every year carries a destination string. A string the country concordance does not recognise is assigned to the comparison arm rather than discarded, so the comparison arm is defined as the complement of the treated one and not as a list.

**The aggregation key is the arm, not the destination.** Records are summed to cells by firm, product, treated-arm indicator and year. Products are carried at the six-digit level and mapped to the 1996 revision of the harmonised system, with codes the concordance does not cover kept as they stand. The arm indicator takes two values, so a cell on the comparison side pools every non-US destination the firm serves in that product and year. We use *cell* throughout for this object, as section 2 does, and section D.5 states what follows from it for the estimating equation.

**Table 12:** From customs records to the estimation sample

| Step                                                   | Units       | Change  |
|--------------------------------------------------------|-------------|---------|
| Customs records, 2000–2014                             | 149,357,863 |         |
| Export records                                         | 105,490,292 |         |
| With a readable six-digit product                      | 105,490,240 | –52     |
| Firm $\times$ product $\times$ arm $\times$ year cells | 44,198,489  |         |
| With both policy regressors                            | 6,863,761   |         |
| Matched to a cell price and quantity                   | 6,817,778   | –45,983 |
| With 2001 input-tariff exposure                        | 6,817,690   | –88     |
| After singleton removal                                | 6,783,323   | –34,367 |
| Estimation sample                                      | 6,783,323   |         |
| On the treated arm                                     | 1,276,498   | 18.8%   |

*Notes:* counts are pooled over 2000–2014. The fifty-two dropped records are worth USD 622,844. A cell carries both policy regressors when the product has a tariff-schedule gap and the firm has a 2001 input-tariff basket; those cells are 15.5% of all cells and carry 25.1% of export value. The per-year build statistics record 6,863,766 cells at that step, and the merge that follows additionally requires strictly positive cell value. Singleton removal is iterative over the three fixed-effect margins — firm  $\times$  year, product  $\times$  year and arm  $\times$  year — and stops when no margin has a group of one. The estimation sample covers 26,142 firms and 4,136 products. The treated-arm share is the share of cells; on the cells carrying both policy regressors the treated arm holds 18.2% of export value.

## D.2 The transaction rules behind the cell price and quantity

Section 2.2 states in three sentences how a cell price is formed. This is the full rule, with the counts each step removes.

**Quantity.** The source reports a value and a value per physical unit. The quantity is their ratio wherever the value per unit is strictly positive, and is missing otherwise. A transaction with no usable quantity enters neither the numerator nor the denominator of eq. (1): excluding it from the quantity sum alone would bias the cell price upward. Over the fifteen years 637,184 transactions are set aside this way.

**One unit per product-year.** Quantities can be added only if the lines share a physical unit. Within each product and year we retain the transactions recorded in the unit code that accounts for the most export value in that product-year, and drop the others: 905,967 transactions worth USD 105.4 billion, which is 0.6% of the export ledger. The share is uneven across years — below 0.25% of records in every year through 2006 except 2005, where it is 1.18%, and between 0.88% and 1.33% in each year from 2007 — and we disclose it rather than act on it. In a small number of product-years the most common unit code is itself missing, and those transactions are then the ones retained; this reaches 44 of 4,502 product

strata in 2007 and at most two strata in any other year.

**Cells that cannot carry a price.** A cell with no usable-quantity transaction, or with a non-positive quantity, is dropped rather than carried with a missing outcome — 230,608 cells over the window, of which 44,040 fall in 2005. The rule that also drops a cell whose value entering the price is non-positive binds in no year.

### D.3 The exposure measure: provenance and the concordance

**Provenance, of which only the last step is ours.** The column-2 rates originate in the Tariff Act of 1930 as carried in the United States tariff schedule. We hold neither the statute nor the schedule. The file we read is the replication package of Handley and Limão (2017), so the exposure measure is their construction rather than ours, including its treatment of specific duties and its concordance across tariff revisions.

**We did not succeed in tracing the step between the statute and that file.** The package’s documentation refers to the article’s data appendix, which gives a source for the applied rates and none for the column-2 rates. Pierce and Schott (2016), on the same policy, attribute both of their rates to Feenstra, Romalis, and Schott (2002), whose own working paper describes a tariff database built on most-favoured-nation rates and does not mention column-2 rates. We record the gap rather than close it.

**The step that is ours we have verified.** The difference between the two schedule columns reproduces the estimation panel’s regressor at every percentile of table 1, to five decimal places, over the 3,994 codes that appear in both the schedule file and the panel.

**The concordance.** The schedules are published at the 1996 revision of the harmonised system and the customs panel is carried at the 1988/92 revision. The published mapping between them is strictly many-to-one in that direction — 5,112 pairs, no 1996 code mapping to two parents — so carrying the gap across requires an average only where a parent has several children. Of the parent codes, 4,753 have one child and 116 have more; 94 have more than one child carrying a gap, and those are the only products on which the averaging rule can matter. Among them the within-parent standard deviation of the gap averages 0.012 and is exactly zero for over half, against a cross-sectional standard deviation of 0.141. Weighting children by pre-period US imports rather than equally moves the measure by 0.0002 on average. Products the mapping leaves uncovered are systematically thin: covered products carry between 95.9% and 97.9% of US export value in every year of the window, with no break at any revision boundary.

Table 13 reconciles the product counts that appear in section 2 and in table 1.

**Table 13:** Products, from the tariff schedule to the estimation sample

| Step                                                  | HS6 products |
|-------------------------------------------------------|--------------|
| Schedule file, 1996 revision                          | 5,033        |
| With a rate actually applied                          | 4,696        |
| With a column-2 rate                                  | 4,361        |
| With both, so that exposure is defined                | 4,348        |
| Carried to the 1988/92 revision of the panel          | 4,218        |
| In the estimation sample                              | 4,136        |
| In both vocabularies, where the regressor is verified | 3,994        |

*Notes:* the first four rows count codes of the schedule file and the next two count parent codes of the panel, so the two blocks are not nested. Exposure is defined only where both rates are present; the 685 products missing at least one of the two carry no exposure and cannot enter the estimation sample. The last row is the set on which the schedule difference and the panel regressor are compared percentile by percentile, and it is an intersection of two code vocabularies rather than a sample restriction.

## D.4 What the treated arm resolves to

The arm is built from the destination string, not from a destination list. The build translates the customs country string into an ISO country code and sets the treated indicator to one for the United States code. Seven strings in the concordance carry the US code: the United States, Puerto Rico, Guam, American Samoa, the Northern Mariana Islands, and the US Virgin Islands, which the source spells two ways. The same translation defines the treated group in the quota setting, where the destination variable takes three values — the United States, the European Union and the rest of the world — rather than two.

**Five of those seven strings never occur.** Table 14 reports the arm’s composition over the full export ledger. Guam, American Samoa, the Northern Mariana Islands and the US Virgin Islands appear in no export record in any of the fifteen years, under either spelling. The treated arm is the United States and Puerto Rico, and Puerto Rico is 1.5% of its records and 0.2% of its value.

## D.5 What the arm definition does to the design

Because the cell is defined on the arm, the index  $d$  in eq. (3) takes two values and  $\delta_{dt}$  is an arm  $\times$  year effect.

**They absorb everything common to US shipments in a year and everything common to non-US shipments in a year, and nothing finer.** US demand, the dollar exchange rate and any US trade-policy change that is not product-specific are absorbed,

**Table 14:** Composition of the treated arm, customs export ledger 2000–2014

| Destination          | Records   | Share | Value (USD bn) | Share |
|----------------------|-----------|-------|----------------|-------|
| United States        | 7,603,497 | 98.5% | 3,188.3        | 99.8% |
| Puerto Rico          | 114,101   | 1.5%  | 6.0            | 0.2%  |
| Other US territories | 0         | 0.0%  | 0.0            | 0.0%  |
| Treated arm          | 7,717,598 | 100%  | 3,194.4        | 100%  |

*Notes:* the frame is every Chinese customs export record for 2000 through 2014 — 105,490,292 records worth USD 17,535 billion — pooled, before aggregation to cells and before any sample restriction. It is therefore not the estimation sample: the estimation sample carries the arm indicator and no destination string, so the split cannot be computed on it. The third row covers the five remaining strings the concordance maps to the United States: Guam, American Samoa, the Northern Mariana Islands, and the US Virgin Islands under both of the spellings the source uses. Shares are of the treated arm. Value is current USD.

which is what the design requires of them. A shock to a single comparison destination is absorbed only through its weight in the comparison-arm average, and a shock that hits two comparison destinations in opposite directions is not absorbed at all. This is why section 3.2 treats the comparison arm as live rather than inert, and why the quota removal is handled there explicitly rather than assigned to  $\delta_{dt}$ .

**The two arms are not comparable objects at this grain, and the asymmetry runs in the convenient direction.** On the 1,989,260 ledger cells for the two pre-period years, a treated cell contains 1.01 distinct destination strings on average and never more than two; a comparison cell contains 2.12. The treated arm is close to a single market; the comparison arm is an aggregate of markets, and the model’s market-level objects should be read against it with that in mind.

The same binary grain governs the singleton rule. The third fixed-effect margin in table 12 is  $\text{arm} \times \text{year}$ : two groups a year, and in the estimation sample the smallest of the thirty holds 51,949 cells. We do not decompose the 34,367 singleton drops by the margin that triggered them. What the  $\text{arm} \times \text{year}$  effects do not absorb is the estimand: the triple interaction varies across products and firms inside every arm–year cell.

## D.6 The inputs to measured quality are winsorised

The log unit value and the log quantity are trimmed before eq. (2) is formed. **The order is: aggregate the transactions to the cell, take logs, then clip.** Bounds are the first and ninety-ninth percentiles within a product and year, computed unweighted on the cells eligible for estimation and *pooled across the two arms*, so a cell is not compared against its own arm alone. A value outside its bounds is replaced by the bound. **No cell is dropped**

**for lying outside them.** A cell in a product stratum that contains no eligible cell has no bounds and does not enter the sample.

One detail of the percentile sample is worth stating because it cannot be avoided. Singleton removal happens downstream, at estimation, and depends on the merged frame; the percentiles are therefore computed on the eligible cells before singleton removal, which is the closest sample constructible at build time.

On the 6,817,778 cells that reach the merge, 204,670 log unit values and 180,722 log quantities are replaced — 3.0% and 2.7% of the cells respectively. Both outcomes enter  $\ln \lambda$ , so measured quality inherits the replacement on either leg.

## D.7 The fifty-states arm

Restricting the treated arm to the fifty states and the District of Columbia removes Puerto Rico from it. Table 14 bounds what that costs: the arm loses 1.5% of its export records and 0.2% of its export value, whether those records are moved to the comparison arm or dropped. The five remaining territory strings are not in the data, so the restriction reaches nothing else. **It changes the composition of the treated arm and not its definition**, and it cannot change the definition, because the arm is constructed from the destination string by a rule that is the same under either convention.

**We do not report an estimate on the fifty-states arm.** No specification in this paper has been re-run with Puerto Rico outside the treated group. What table 14 establishes is the size of the object at issue and not the sensitivity of any coefficient to it.

## D.8 The quota panel and its sample

Section 2.2 describes the second setting’s panel; this is the count surviving each step of it. The record-level rules are not repeated: the rebuild applies those of section D.2 unchanged, and differs from the main panel only in carrying the destination as three groups rather than two.

## D.9 What the quota estimates rest on

Section 4.4 reports the quota event study, its pre-period and its sensitivity analysis. Three further checks are collected here.

**Not every product in that sample was treated in 2005.** China became eligible for quota removal on joining the World Trade Organization at the end of 2001, and the quotas on goods placed in the earlier integration phases were relaxed in early 2002 rather

**Table 15:** The quota setting, from quota lines to the estimation sample

| Step                                          | Units     | Change  |
|-----------------------------------------------|-----------|---------|
| <i>Panel A. Products</i>                      |           |         |
| Eight-digit lines in the US quota file        | 491       |         |
| Products they reach                           | 442       |         |
| After dropping a missing or zero maximum fill | 424       | −18     |
| <i>Panel B. Cells</i>                         |           |         |
| Panel, three destination groups, 2000–2008    | 3,823,874 |         |
| United States                                 | 394,096   | 10.3%   |
| European Union                                | 721,797   | 18.9%   |
| Rest of the world                             | 2,707,981 | 70.8%   |
| After dropping the European Union             | 3,102,077 |         |
| After singleton removal                       | 3,047,220 | −54,857 |
| Estimation sample                             | 3,047,220 |         |

*Notes:* the panel covers 94,323 firms, 424 products and nine years. Of the 424 products, 181 have a quota child filled above 0.90. A product whose maximum fill rate is missing or exactly zero is dropped from both arms rather than assigned to either, because its pre-period status is ambiguous. Singleton removal is iterative over firm  $\times$  year, product  $\times$  year and destination-group  $\times$  year, as in table 12; no cell is lost to a missing outcome, so the whole of the 54,857 is singleton removal. The final count is the  $N$  of the event study in section 4.4.

than on the final date (Khandelwal, Schott, and Wei, 2013). Five of the quota categories our products map into are of that kind. Dropping every product with a line in one of them removes 30 of 424 products, a tenth of the cells and 7.8 percent of the value, and it moves nothing. Figure 2 A plots both paths and they lie on top of one another: the average post-period coefficient goes from  $-0.686$  to  $-0.682$ , the yardstick from  $0.406$  to  $0.404$ , and the sensitivity analysis, re-estimated on the smaller sample, returns the same average breakdown value. **The mis-dated products are in the sample and they are not what produces the estimate**, and dropping them puts the rest on the footing Khandelwal, Schott, and Wei (2013) work from.

**Three of the four post-period years are not open-trade years either.** From 1 January 2006 to the end of 2008 the United States held a list of Chinese categories back under agreed import levels (section 2.1), and 18 of the 75 quota categories our products map into are on it. Dropping every product with a line in one of them removes 80 of 424 products, a fifth of the cells and a quarter of the value — and unlike the previous check, this one moves the estimates. The post-period average falls from  $-0.686$  to  $-0.586$ , the largest pre-period step rises from  $0.406$  to  $0.438$ , and the breakdown values fall with them, to  $0.65$ ,  $0.30$ ,  $0.40$  and  $0.15$  by post-year and  $0.30$  for the average.

**That exclusion is not a cleaning operation, and the direction of the movement is what says so.** Had the re-imposed limits merely undone part of the liberalisation, the safeguarded products would carry the smaller declines and dropping them would make the estimate larger. It does the opposite. Safeguards were imposed where imports surged, which is the statutory trigger, so the products removed are the ones the shock landed on hardest; Bloom, Draca, and Van Reenen (2016) decline to use post-2005 quota levels in their own design for that reason. **The smaller estimate on what remains is the effect with the largest responses taken out — a lower bound rather than a cleaner number**, and we report it as one. Khandelwal, Schott, and Wei (2013) report their own results unchanged under this exclusion; ours move, in the direction that selection implies.

**The pre-period earns less than it appears to, and we state how much.** A joint test of the pre-coefficients divides on whether the first year is included — excluding it the test does not reject, including it the test rejects — and a test that fails to reject settles nothing unless it could have detected a violation that mattered (Roth, 2022). So we computed what ours could detect. Against a linear violation large enough to account for the whole estimated post-period decline, both tests reject with probability indistinguishable from one: **the pre-period does rule out large linear divergence, and that is a real thing to know**. Against one accounting for a quarter of it the same tests reject with probability  $0.28$  and  $0.48$ , so a violation of that size would be missed more often than not.

**And against a counterfactual shift dated at the treatment itself, the power of any pre-test equals its size:** such a violation leaves every pre-period coefficient untouched, so no pre-period evidence of any kind can speak to it. That last case is the reason a sensitivity analysis is not optional here, and it is why table 6 rather than fig. 2 A carries the weight.

One feature of that test bears stating. **The breakdown values depend on how far back the window reaches, and ours reaches further back than this literature’s.** We estimate over 2000 through 2008. Khandelwal, Schott, and Wei (2013), studying the same quota removal, work in two overlapping three-year windows that together span 2002 to 2005: their difference in differences compares the change from 2004 to 2005 against the change from 2003 to 2004, and their pre-reform check compares 2003 to 2004 against 2002 to 2003. The pre-period movement that drives our breakdown values sits between the first two years of our window and therefore lies outside theirs entirely. Because the yardstick in  $\bar{M}$  is the largest violation between consecutive pre-periods, a shorter window would shrink that yardstick and return more favourable numbers than table 6 reports. **We report the longer window, which is the one we set before computing the sensitivity, and we report what it gives.** Which window a reader should prefer is a question about how far back comparability extends in this panel, and we do not settle it here.

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